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Chaotic mixing performance of non-Newtonian fluids in S-Type static mixer and optimization based on response surface methodology

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ABSTRACT

Non-Newtonian fluids play a vital role in industrial production. It is of great significance to study the mixing and chaotic characteristics of flow field in static mixers for the energy utilization. An improved structure of S-Type static mixer is proposed. In order to deeply investigate the flow characteristics of non-Newtonian fluids in S-Type static mixers, experimental and numerical studies are conducted on the mixing efficiency and chaotic behaviors under different flow rates. The PLIF method is adopted to visually study the distribution of CMC solutions with different mass concentrations. It is found that the average deviation of coefficient of variation (CoV) between experiment and Second-order simulation scheme is 4.15 %. The simulation results show that the *M number* in S-Type-b mixer is 10.13–18.29 %, 10.82–22.26 %, and 20.39–27.35 % higher than that of the Kenics, Komax, and S-Type static mixers, respectively. Additionally, the S-Type-b static mixer exhibits a higher average Lyapunov exponent (LE_{ave}) and lower variance of the Lyapunov exponent (LE_{var}) under different flow rates, which indicates that the S-Type-b static mixer has better flow stability. The structural parameters of S-Type-b mixer are optimized using Response Surface Methodology (RSM) and Non-dominated Sorting Genetic Algorithm II (NSGA-II). The optimal design parameters ensure that the *M number* of the S-Type-b static mixer is 0.2926 and $\overline{LE}_{ave}$ reaches 5.3246, which significantly improves the mixing efficiency and system stability.

1. Introduction

Mixing is a crucial process in modern industrial applications [1], particularly in the fields of chemical production [2], food processing [3], pharmaceutical [4] and environmental engineering [5]. Because of the rheological characteristics of some high-viscosity chemicals, foods and biological materials, it is difficult to achieve ideal mixing effect using traditional mixing methods [6]. When the fluid media such as proteins, polymer materials, fibers and cell tissues are sensitive to shear stress, mixing operation is often completed in laminar flow conditions because high-speed shear motion would destroy the macromolecular chains. However, reaction rates are affected by the poor mixing in laminar flow states with low Reynolds numbers (*Re*), which would reduce the production efficiency and increase the waste of raw materials [7]. To effectively solve these problems, it is very significant to achieve chaotic mixing because it promotes fluid deformation through multiple stretching, cutting and folding to enhance the diffusion rate within laminar flow. Meanwhile, it also improves momentum and mass transfer to achieve more efficient mixing and homogenization [8–11]. Static mixers are widely concerned because of their simple structure, easy maintenance, high energy efficiency and excellent mixing performance.

As an efficient mixing device in chemical process, static mixers could meet the mixing requirements from single phase to multiphase fluids, covering a variety of systems such as liquid−liquid [12–14], gas−liquid [15–18] and solid−solid [19–21]. It also shows remarkable performance in dealing with the mixing of Newtonian and non-Newtonian fluids [22–25]. The mixing performance of Newtonian fluids in static mixers is mainly related to parameters such as flow velocity and *Re*. However, non-Newtonian fluids such as shear thinning, shear thickening and viscoelastic fluids have complex rheological behaviors, which have a more significant effect on the flow field, shear action and energy consumption in the mixer. The mixing performance of Kenics static mixer under diverse operating conditions has been widely studied. The mixing efficiency of non-Newtonian fluids in Kenics mixer was more than twice that of Newtonian fluids at low *Re*, while the mixing efficiency was only about 15 % higher that of Newtonian fluids at $Re = 1000$ [22]. The addition of high viscosity fluids to low viscosity fluids would cause a low

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Nomenclature

Abbreviations

ANOVA	Analysis of variance
CMC	Carboxymethyl cellulose
CoV	Coefficient of variation
ERT	Electrical resistance tomography
GA	Genetic algorithm
MSE	Multi-stacked elements
MI	Mixing index
NSGA-II	Non-dominated sorting genetic algorithm II
PLIF	Planar laser-induced fluorescence
RSM	Response surface methodology
A	Cross-sectional area [m^2]
A_r	Aspect ratio
a_i	Area of the ith cell in the cross-section [m^2]
C_{i1}	Concentration of the main pipe fluid [wt%]
C_{i2}	Concentration of the injection pipe fluid [wt%]
c_i	Mass fraction of the *i*th cell
$\bar{c}$	Average mass fraction of the injection pipe fluid
D	Main pipe diameter [m]
d	Injection pipe diameter [m]
f	Friction factor
g	Gravitational acceleration
$\boldsymbol{I}$	Stretch vector of particles
$\boldsymbol{I}_0$	Initial stretch vector of particles
K	Consistency coefficient of the power-law fluid [Pa·s^n]
K'	Consistency coefficient of a generalized non- Newtonian fluid [Pa·s^n]
LE_{ave}	Average Lyapunov exponent
LE_{var}	Variance of the Lyapunov exponent
$\overline{LE}_{ave}$	Average value of average Lyapunov exponent
l	Blade length [m]
l_m	Mixing section length [m]
l_o	Distance from the last element to the outlet [m]
l_2	Length of inlet 2 [m]
N	Total number of cells
n	Power-law index
P_{in}	Pressure at the inlet of the static mixer [Pa]
P_{out}	Pressure at the outlet of the static mixer [Pa]
Δp	Pressure drop [Pa]
$\Delta p_{empty\ pipe}$	Pressure drop in empty pipe [Pa]
Q_{i1}	Flow rates of the main pipe fluid [L/min]
Q_{i2}	Flow rates of the injection pipe fluid [L/min]
Q_r	Flow ratio
R^2	Determination coefficient
R^2_{adj}	Adjusted coefficient of determination
R^2_{pre}	Predicted coefficient of determination
$T_{z/l}$	Average time taken for all particles to traverse the z/l element [s]
$\boldsymbol{v}$	Velocity vector
$\bar{v}_z$	Average axial velocity [m/s]
W_{i2}	Mass fraction of the injection pipe fluid
z	The axial position [m]
Z_f	Z factor

Greek symbols

α	Extensional efficiency
γ	Deformation rate tensor [s^{-1}]
$\dot{\gamma}$	Shear rate [s^{-1}]
δ	Blade thickness [m]
λ_k	Stretch rate of the kth particle
μ_a	Apparent viscosity [Pa·s]
ζ_M	Mixing efficiency parameter
$\zeta_{M,out}$	Mixing efficiency at the outlet of the static mixer
$\nabla \boldsymbol{v}$	Velocity gradient
ρ	Fluid density [kg/m^3]
σ	Standard deviation of concentration
σ_{max}	The maximum standard deviation of concentration
τ	Stress tensor [Pa]
ω	Vorticity tensor [s^{-1}]

Dimensionless Group

Re	Reynolds number
Re_g	Generalized Reynolds Number

mixing intensity of about 20 % [26]. The *MI* of mixers could almost reach 0.99 when the power-law index (n) was equal to 0.49 – 1, while the pressure drop (Δp) increased with the rise of Re and n [27]. The decline of mixing performance was primarily because of the asymmetric flow at the intersection of Kenics mixing elements [28]. In comparison with the Kenics mixer, SMX static mixer had better mixing effect in non-Newtonian fluids and Newtonian fluids. After the last SMX element, the Δp and CoV of the shear thinning fluid were lower by 2.119 MPa and 0.51 than those of Newtonian fluids, respectively. It indicated that shear thinning fluids had higher mixing efficiency [29]. As the fluid velocity ratio increased from 0.6 to 171.5, the power consumption decreased from 19.50 kW to 1.24 kW [30]. When the blade angle of SMX elements was 45°, the Δp and CoV were the lowest with the thickness of 1 mm and eight blades [31]. The Δp in Komax mixer increased with the rise of flow rates, the friction factor (f) reduced as increased Re and the amplitude of this decline was large in laminar flow state [32]. The mixing performance of MSE mixer with 0.75 wt% and 1.50 wt% CMC solution was the best and its mixing rate was 96.30 % and 92.55 %, which was significantly better than Kenics mixer and empty pipe [33]. The mixing efficiency of Kenics mixer would not be obviously improved with the rise of Re, while the shear thinning fluid in SMB-R mixer fluctuated strongly at a lower Mach number [34]. The SMX_N static mixer improved the mixing quality because of the presence of broken wave structure when the Δp was constant [35]. Table 1 summarizes the research on the mixing performance of a variety of static mixers (including Kenics, SMX, Komax, etc.) in dealing with Newtonian fluids and non-Newtonian fluids.

Previous studies predominantly focused on the effects of different fluid characteristics on the mixing efficiency and Δp in a single static mixer. In the comparative study of multiple static mixers, multi-layer designs (such as SMX) usually have the advantage of efficient mixing, but it is accompanied by higher Δp [31,35,55]. As a result, it is necessary to design an efficient static mixer with lower Δp. Although the S-Type static mixer as a widely used mixer in Statiflo Company shows efficient mixing performance in high-viscosity fluids, the research of S-Type static mixer in non-Newtonian fluids mixing is still insufficient. Therefore, an improved S-Type static mixer (S-Type-b) is proposed in this study, which can reduce the flow resistance of non-Newtonian fluids and improve the mixing performance by removing the side baffle. The mixing characteristics and chaotic behaviors of two blade-type static mixers (S-Type and S-Type-b) are compared with other mixers (Kenics and Komax) for the first time, which enriches the mixing mechanisms of static mixers in complex fluids. Firstly, the geometric structures of four mixers as well as the experiment and simulation schemes are introduced. The validity verification of the model is supported by visual analysis using PLIF. Secondly, the mixing efficiency and chaotic characteristics of four mixers are comprehensively quantified and evaluated by multidimensional parameters such as Z_f, mixing index (*MI*), *M number*,

Table 1
The study of static mixers in Newtonian fluids and non-Newtonian fluids.

Static Mixers	Researches	Operating condition	Fluid media	Methods	Evaluation indexes
Kenics	Rahmani et al. [22] (2005)	Re = 0.01–1000	50–5000 ppm CMC solution; Newtonian fluids	Discrete phase model; Carreau model; Power-law model	PDU; Z_f; Mixing efficiency
	Alberini et al. [26] (2014)	Re = 88–91	80 wt% glycerol solution; 0.1 wt% and 0.2 wt% Carbopol 940 solutions	PLIF	CoV; Δp; Mean value of grayscale; Striation thickness; Log variance
	Mahammedi et al. [27] (2021)	Re = 0.1–500	0–0.7 wt% CMC solution	Species transport model; Laminar flow model; Power-law model	MI; Δp; Standard deviation of mass fraction
	John et al. [28] (2024)	Re = 10–30	Viscoelastic fluids	FENE-CR fluid model; Laminar flow model	Degree of asymmetry; Striation thickness
SMX	Liu et al. [29] (2006)	Re = 0.001–100	Shear thinning fluids	Discrete phase model; Power-law model; Laminar flow model	CoV; Δp
	Jegatheeswaran et al. [24] (2018)	Re = 0.1–1.26	0.5 wt% xanthan gum solution; Water	Species transport model; Herschel-Bulkley model; ERT	CoV; Δp; Power consumption
	Gayen et al. [31] (2024)	Re = 10 and 100	Newtonian fluids and non-Newtonian fluids	Discrete phase model; Power-law model	CoV; Z_f
Komax	Revathi et al. [32] (2020)	Q = 14–26 m^3/s	Xanthan gum solution; Starch solution	Pressure measurement	Z_f; f
Kenics & MSE	Mochizuki et al. [33] (2017)	Re = 11,700	Glycerin solution; 0.75 wt% and 1.50 wt% CMC solution	PLIF	Mixing rate; Mean value of grayscale
Kenics & SMB-R	Migliozzi et al. [34] (2021)	Re = 0.6 and 20	Glycerol solution; Polyethylene glycol solution; 200 ppm polyacrylamide solution; Xanthan gum solution	Species transport model; Carreau model; multi-mode Maxwell model; PLIF	CoV; f; PSD
Inliner series 45 & Komax & SMX_M & SMX_N	Moghaddami et al. [35] (2023)	Re = 0.1–150	CMC solutions	Species transport model; Power law model; Laminar flow model	CoV; Δp

extensional efficiency (α), and Lyapunov exponent under different flow rates. Finally, RSM and NSGA-II are utilized to optimize the S-Type-b mixer to ensure the stability of chaotic mixing behavior and improve the mixing efficiency, which provides important theoretical instruction for the design and performance optimization of new static mixers.

2. Experiment

2.1. Experimental devices

CMC aqueous solutions, as a commonly used non-Newtonian fluids in industry, are utilized as the fluids in main pipe and injection pipe in this study. The solution is prepared with the same CMC powder as studied by Sun et al. [36], and the rheological properties were based on the data of this study. In the experiment, the CMC mass concentration of the fluids in main pipe and injection pipe is 0.5 wt%, in which rhodamine B is added into the injection pipe fluid as a tracer and excited by a semiconductor laser with a wavelength of 520 nm and an output power of 1 W. The temperature of all fluids remains at 25 °C throughout the experiment to ensure consistency and reliability of the results. The PLIF visualization experimental setup includes a mixing apparatus and an image acquisition system as depicted in Fig. 1. The four Kenics elements in the mixing apparatus are staggered by 90° and the rotation directions are opposite. The specific geometric parameters of Kenics are detailed in Section 3.2. These elements are placed in a plexiglass pipe with an inner diameter of 40 mm and made by 3D printing using transparent resin materials. A jacket is added to the outside of the plexiglass pipe for a clearer image. A self-priming centrifugal pump and peristaltic pump are adopted to deliver the fluids in main pipe and injection pipe, respectively. The flow rate of the main pipe fluid (Q_{i1}) is controlled by a digital flowmeter and the peristaltic pump precisely controls the flow rate of the injection pipe fluid (Q_{i2}). The fluid mixing process is captured by the image acquisition system adopting a high-speed camera. A filter is installed in front of the camera to filter out non-fluorescent light signals. The key technical parameters of the experimental devices are presented in Table 2.

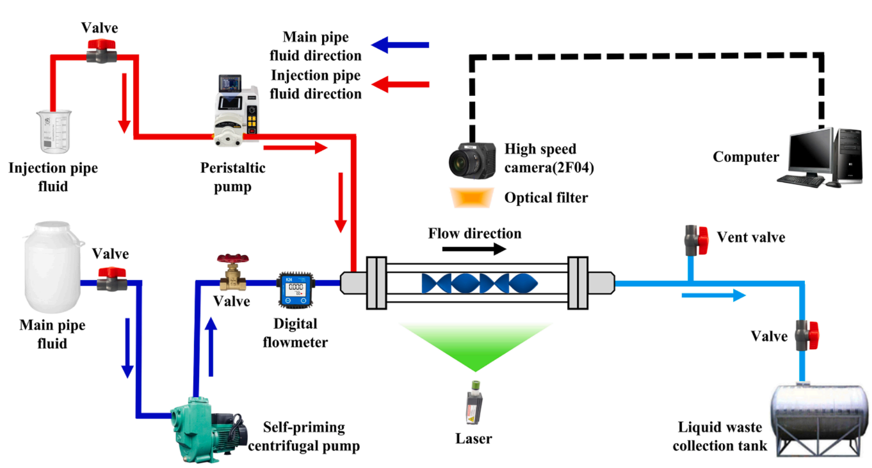

Fig. 1. PLIF visualization experimental device.

Table 2
Key technical parameters of the experimental devices.

Equipment	Model	Range	Accuracy	Company
Semiconductor laser	AJHXB520XXXD	12–24 V	±1 %	Zhuhai Maizhi Laser Co., Ltd.
Self-priming centrifugal pump	25WBZ4–13	0–66.67 L/min	±0.7 %	Haicheng Three Fish Pump Industry Co., Ltd.
Peristaltic pump	BT-600CA/ 253Yx-PPS	1.76×10^{-4}–3.365 L/min	±2 %	Chongqing Jieheng Peristaltic Pumps Co., Ltd.
Digital flowmeter	K24	0.1–8 L/ min	±0.4 %	Wenzhou Lanbao Technology Co., Ltd.
High-speed camera	Revealer-2F04M	1920 × 1080 pixels	152 μm × 148 μm	Hefei Fuhuang Junda High-tech Information Technology Co., Ltd.

2.2. Experimental approach

The CoV is a key index for evaluating the mixing quality of static mixers, which can be calculated by concentration distribution on different cross sections to quantify the uniformity of fluid concentration [26,31,35,37,38]. When the CoV value is less than or equal to 0.05, it is generally considered to have reached a fully mixing state [39–41]. The definition of CoV is as follows [42]:

$$\bar{c} = \frac{1}{N}\sum_{i=1}^{N} c_i \tag{1}$$

$$\sigma = \sqrt{\frac{1}{N}\cdot\sum\nolimits_{i=1}^{N}(c_i - \bar{c})^2} \tag{2}$$

$$\mathrm{CoV} = \frac{\sigma}{\bar{c}} \tag{3}$$

where N represents the total number of cells, $\bar{c}$ represents the average mass fraction of the injection pipe fluid and c_i denotes the mass fraction of the injection pipe fluid in the ith cell.

The captured original image in the experiment is a grayscale image, because it only contains the intensity information of light. As a result, the fluorescence distribution comparison is poor to make it difficult to distinguish differences in fluorescence intensity. The operation steps in Fig. 2 are performed by using MATLAB to enhance the contrast and clarity of the images. In this study, the grayscale image from Choi et al. [43] is input into the image processing program of this study, and the generated pseudocolor effect is compared with the original pseudocolor image to verify the reliability of program. Fig. 2(a) and (g) are the original grayscale image and pseudocolor image of this study, and Fig. 2 (f) is the pseudocolor image processed by the MATLAB program in this study. The comparison results show that the CoV deviation between Fig. 2(g) and (f) is only 1.05 %, which verifies the accuracy and consistency of MATLAB image processing program.

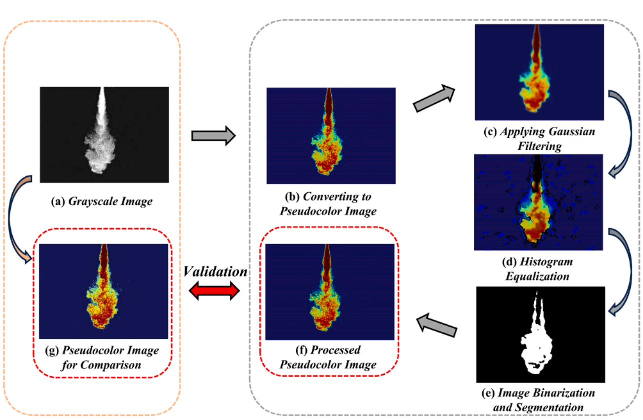

Fig. 2. Flow chart of MATLAB processing pseudocolor image.

The uncertainty analysis method is used to calculate the maximum allowable error of the measurement parameters and devices in the experimental system [44].

$$U_R = \sqrt{\sum\nolimits_{i=1}^{n}\left(\frac{\partial R}{\partial x_i}U_{x_i}\right)^2} \tag{4}$$

where U_{x_i} is the standard uncertainty of a single error source. In this experiment, the uncertainties of Q_{i1} and Q_{i2} are lower than 0.806 % and 2 %, respectively. The effect of laser intensity fluctuation is approximately ±1 %, and the error range of the pseudocolor image in the image processing program is ±1.05 %. The U_R is controlled within ±2 %, which confirms that the influence on the key experimental parameters is small.

3. Numerical simulation

3.1. Governing equations

The steady-state conservation equation of incompressible fluid in laminar flow state is numerically solved by ANSYS FLUENT 2020 based on the finite volume method. To simulate the flow in the static mixer, the continuity, momentum and species transport equations are solved:

$$\nabla\cdot(\rho\boldsymbol{v}) = 0 \tag{5}$$

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{v}\cdot\nabla\boldsymbol{v}\right) = -\nabla p + \nabla\tau + \rho\mathbf{g} \tag{6}$$

$$\nabla\cdot(\rho\boldsymbol{v}c_{i2}) = \nabla\cdot(\rho D_{i2}\nabla c_{i2}) \tag{7}$$

where ρ is the fluid density, $\boldsymbol{v}$ is the velocity vector, t is the time, p is the static pressure, τ is the stress tensor, $\mathbf{g}$ is the gravitational acceleration, c_{i2} is the local mass fraction of the injection pipe fluid and D_{i2} is the mass diffusion coefficient of the injection pipe fluid with a value of 1×10^{-11} m^2/s. According to previous studies, the mass diffusion coefficient can effectively describe the diffusion characteristics of CMC solutions in the range of $n = 0.49 - 1$ under $Re = 0.1 - 120$ [27,45].

According to the literature of CMC solution with the same concentration as this study, the experimental results have obvious shear thinning characteristics and conform to the power-law model, which do not show yield-stress [36]. A power-law model is employed to predict viscosity change in fluid flow, where viscosity is defined as [46–48]:

$$\tau = K\dot{\gamma}^n \tag{8}$$

$$\mu_a = K\dot{\gamma}^{n-1} \tag{9}$$

where τ is the shear stress, μ_a is the apparent viscosity, $\dot{\gamma}$ is the shear rate, n is the power-law index and K is the consistency coefficient of the power-law fluid.

3.2. Geometry model and fluid properties

The geometry models of different static mixers used for CFD are illustrated in Fig. 3. These mixers, namely Kenics, Komax, S-Type and S-Type-b, are designed for efficient mixing of fluids within a pipe system. The Kenics static mixer consists of a series of helical blade elements arranged alternately at 180°. The Komax static mixer is composed of four

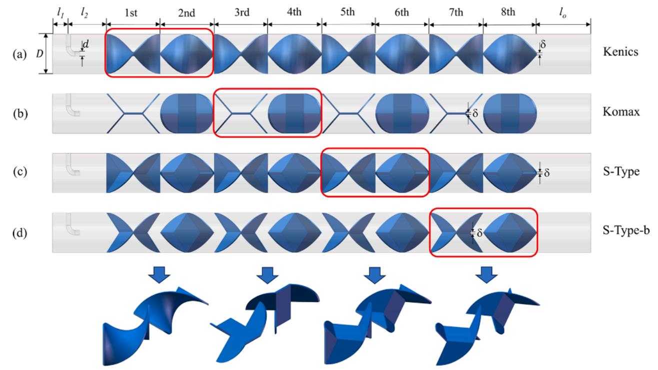

Fig. 3. Geometry models of four different static mixers: (a) Kenics, (b) Komax, (c) S-Type and (d) S-Type-b.

quarter oval plates and one rectangular plate. The S-Type static mixer is constructed with a rhombic plate as the middle plate and two vertically placed triangular side baffles whose height equals to the short diagonal length of the rhombic plate. Finally, two triangular plates are connected diagonally. The S-Type-b static mixer is an improved version of the S-Type, which removes side baffles. The injection pipe is located in the center of the main pipe. The fluid in the injection pipe immediately contacts with the fluid of the main pipe after flowing out and is rapidly cut by the mixing element to mix with the fluid in the main pipe.

The detailed dimension of these static mixers is listed in Table 3, which provides essential data for accurately establishing and interpreting CFD models. The main pipe diameter (D) of 40 mm is mainly considered to match with the experimental devices, which is widely applied in the chemical field [49–51]. In addition, D and δ have been used in relevant numerical simulation studies [52–54], which verify its rationality in meeting the calculation accuracy and optimizing the calculation efficiency. The dimensions of the main and injection pipe are designed based on the study of Haddadi et al. [55]. Aspect ratios (A_r) of 1, 1.5 and 2 are selected to systematically evaluate the influence of geometric complexity on mixing efficiency, chaotic behavior and energy consumption of fluids [56,57]. In Section 4.1, the effects of geometric structure on mixing efficiency and chaotic characteristics are analyzed at $A_r = 1.5$. The length of the empty pipe before and after the mixing element could ensure that the flow is in a fully developed state and prevent reversed flow during the simulation. The distance from the first element to the inlet is equal to the distance from the last element to the outlet (l_o).

3.3. Mesh generation and boundary conditions

The polyhedral mesh is employed to discretize the entire computational domain in Fluent Meshing due to the complexity of static mixer elements. Polyhedral mesh offers higher computational efficiency and accuracy than tetrahedral and hexahedra meshes.

The coupling of velocity and pressure is solved by utilizing the SIMPLEC algorithm. The SIMPLEC algorithm can solve the steady-state problem by optimizing the speed correction coefficient and it also could maintain rapid convergence and reduce the demand for computing resources [50,58]. A second-order upwind discretization approach is employed to solve the concentration and momentum equations. The second-order upwind scheme has higher accuracy than the first-order upwind scheme, especially in high gradient areas. The efficiency of pressure-velocity coupling solution could be improved combined with the SIMPLEC algorithm [59,60]. The convergence of the numerical results is achieved when residual quantities reach 10^{-5}. The outlet boundary condition is defined as pressure-outlet and the wall boundary condition is set as the no-slip condition. The pressure outlet condition allows the fluid to freely flow out of the system. User Defined Function (UDF) is utilized to set different inlet velocities for power-law fluids [61]:

$$v = \frac{3m+1}{m+1} \cdot \left(1 - \frac{\sqrt{x^2+y^2}}{D/2}\right)^{\frac{1}{m}} \bar{v}_z \tag{10}$$

where $m = 0.6$, x and y are coordinate values, $\bar{v}_z$ is the average axial velocity, respectively. The Q_{i1} in the simulation ranges from 0.25 to 2.5 L/min and the Q_{i2} accounts for 80 % of the Q_{i1}. $\bar{v}_z$ can be calculated by the following relationship:

$$\bar{v}_z = \frac{Q}{A} \tag{11}$$

where A denotes the cross-sectional area.

3.4. Gird independence test and model validation

The polyhedral mesh could maintain high accuracy and reduce the consumption of computational resources in CFD simulation because of its fewer cells and higher geometric adaptability. During the grid independence test, the cell number are set at 935,026, 1,155,419, 1,363,561, 1,679,343, 2,105,165 and 4,366,642, respectively. These tests are conducted under the flow conditions of $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min. It can be analyzed from Fig. 4(a) that when the cell number reaches 2,105,165, the CoV and mass fraction of the injection pipe fluid (W_{i2}) exhibit the smallest deviation of 2.2 % and 0.12 %, respectively. This demonstrates that this cell number keeps higher accuracy and also provides a lower computational cost. The Fig. 4(b) shows the velocity distribution along the x-axis for different grid numbers, showing an M-shaped symmetry. Taking W_{i2} of gird 6 as the reference value, the ranges of W_{i2} of other grids are 2.973–51.048 %, 1.058–33.387 %, 0.771–21.486 %, 0.425–19.490 % and 0.273–12.168 %, respectively. Therefore, a mesh with 2,105,165 cell number is selected for subsequent simulation to optimize performance and efficiency.

3.5. Optimization methods

3.5.1. Optimization procedure

RSM is an advanced mathematical modeling technique for optimizing experimental design [62]. The Genetic Algorithms (GA) could effectively search for the global optimal solution by simulating the natural selection processes thus showing higher efficiency and accuracy in the optimization of multi-variable and complex parameter interaction. Among these, NSGA-II exhibits better performance in solving the two-objective optimization [63]. The flowchart of RSM and NSGA-II is depicted in Fig. 5. RSM process includes determining design variables and objective functions, then an optimal design is implemented to obtain the experimental scheme. The analysis of variance (ANOVA) is used to check the significance of the model. If the model is found to be significant, the response surface analysis is carried out, otherwise reevaluate the design. In the NSGA-II process, the population is first initialized and the objective functions are calculated to obtain the Pareto-optimal scheme.

Table 3
Geometrical parameters of static mixers.

Structure parameter	Dimension
Main pipe diameter, D (m)	0.04
Injection pipe diameter, d (m)	0.005
Blade thickness, δ (m)	0.002
Distance from the last element to the outlet, l_o (m)	0.06
Mixing section length, l_m (m)	0.48
Aspect ratio, A_r	1, 1.5, 2

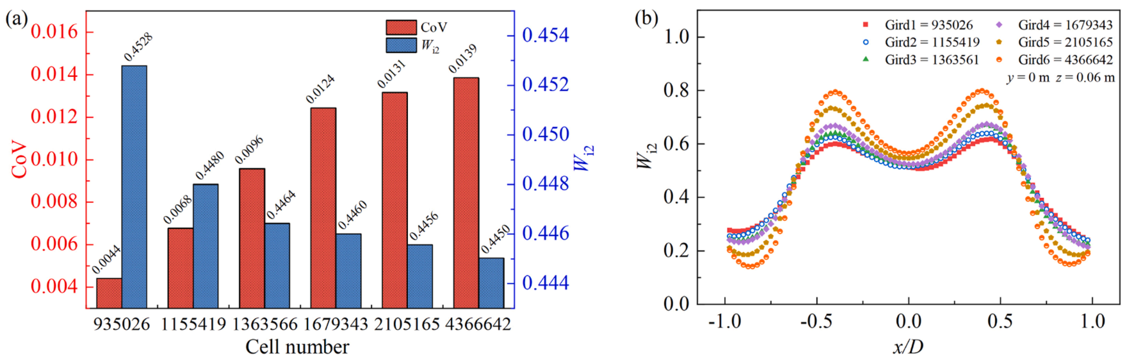

Fig. 4. Grid independence test: (a) Variation of CoV and W_{i2} at $z/l = 8$ and (b) W_{i2} under $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min at $z = 0.06$ m.

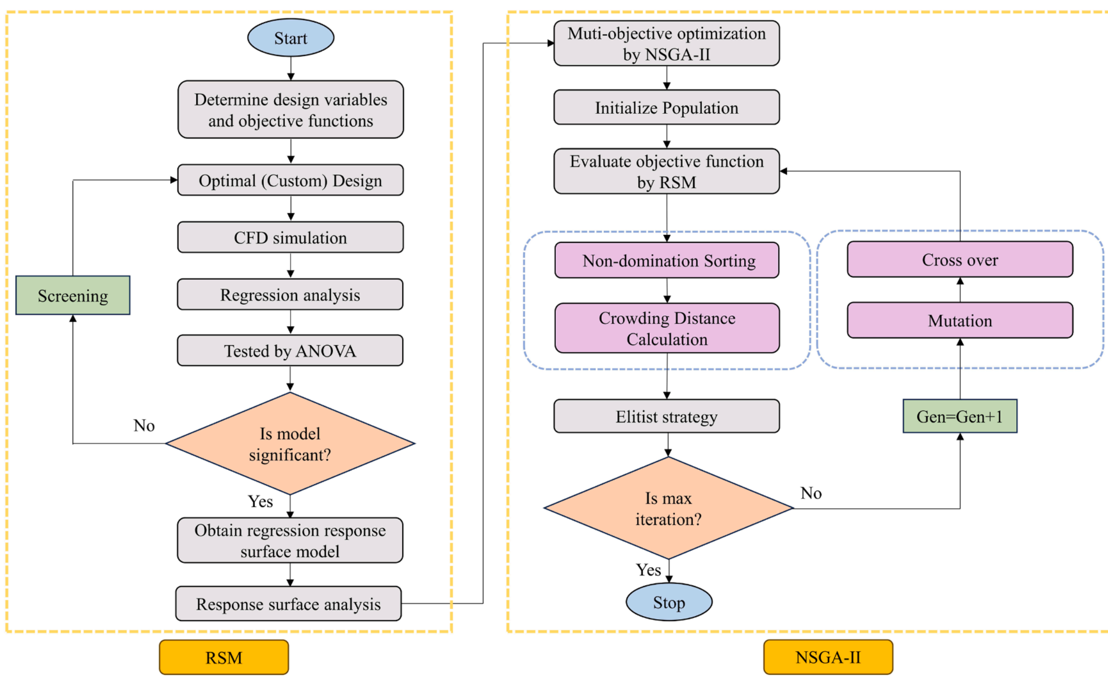

Fig. 5. Flowchart of RSM and NSGA-II.

3.5.2. Response surface methodology

RSM utilizes statistical and mathematical methods to establish a regression model connecting design variables and objective functions. The correlation among design variables (X_1, X_2, ……, X_k) and the objective function G is represented by this response surface model.

$$G = f(X_1, X_2, \ldots, X_k) + \varepsilon \tag{12}$$

where $f(X)$ is the desired target, also called the approximate function, and ε represents the composite error. The most common mathematical model form of response surface objective functions is low-order polynomials, including first-order or second-order models. When the first-order model fails to adequately capture the interaction between dependent variables and the surface curvature, the second-order model could fit the data well. The typical second-order model equation is as follows:

$$G = \beta_0 + \sum_{i=1}^{k} \beta_i X_i + \sum_{i=1}^{k} \beta_{ii} X_i^2 + \sum_{i<j}^{k} \beta_{ij} X_i X_j + \varepsilon \tag{13}$$

where β_0 is the intercept term, $\beta_i X_i$, $\beta_{ii} X_i^2$ and $\beta_{ij} X_i X_j$ is the coefficient for the first-order term, second-order term and interaction term, respectively. In this way, the second-order model can effectively capture the nonlinear relationship and interaction among variables.

3.5.3. ANOVA

The significance test must be conducted before applying the response surface function to ensure its accuracy and reliability. The determination coefficient (R^2) is an important evaluation index for assessing the model fitting degree. The Eq. (14) for calculating R^2 is as follows:

$$R^2 = 1 - \frac{SS_E}{SS_T} = \frac{SS_R}{SS_T} \tag{14}$$

where SS_E denotes the error sum of squares, SS_R represents the regression sum of squares, and SS_T stands for the total sum of squares. The R^2 is closer to 1, indicating that the model is more accurate. However, R^2 may also increase with the increase of polynomial fitting degree, which leads to overfitting and reduced model reliability. Therefore, the adjusted determination coefficient (R^2_{adj}) is introduced, which offers a more stable evaluation of the model accuracy. The R^2_{adj} is expressed as follows [64]:

$$R^2_{adj} = \frac{SS_E}{df_e} \Big/ \frac{SS_T}{df_T} \tag{15}$$

where df_e is related to the degrees of freedom for the error term, while

df_T corresponds to the comprehensive degree of freedom for the error term. The value of R^2_{adj} is always limited between 0 and 1, indicating a better fit as it approaches 1. It is generally observed that R^2_{adj} is lower than R^2, but when R^2_{adj} is significantly less than R^2, it suggests that there may be redundant terms in the polynomial, which means that the model requires to be refitted.

3.5.4. NSGA-II algorithm

As a genetic algorithm for multi-objective optimization, NSGA-II has demonstrated its effectiveness and applicability in dealing with complex calculation [65]. In contrast to the traditional method, the Pareto-optimal solutions are obtained by NSGA-II in a single run, which greatly reduces time and computational resources. The effectiveness of the algorithm derives from the application of non-dominated sorting and crowding distance function. It not only identifies the optimal solution at different front layers, but also ensures the uniform distribution of the solution, which prevents clustering in the objective space and maintains population diversity.

4. Results and discussion

4.1. Model validity verification and sensitivity analysis

The validity and sensitivity analysis of CFD model are confirmed by comparing the numerical results with experimental measurement data as illustrated in Fig. 6. The experimental and simulation values present a consistent upward trend as the flow rate increases as illustrated in Fig. 6 (a). The deviation between the simulation results and experimental values of the Second-order method is always <6 %, the maximum deviation is 5.08 % and the average deviation is 4.15 %. In contrast, the Quick method has a slightly higher deviation with a maximum deviation of 5.99 % and an average deviation of 5.07 %, which could stem from the uncertainty of the experimental setup and small fluctuations in the experimental operation. In addition, the deviation of the first-order method significantly increases and the highest reaches 42.41 %, which indicates that the numerical accuracy of this method is low and it is not suitable for describing the concentration distribution of non-Newtonian fluids under laminar flow conditions. As shown in Fig. 6(b), the second-order method is adopted to verify the experimental and simulation results under various injection tube fluid concentrations (C_{i2}). The results demonstrate that the deviation between the simulation value and experimental value is <6 % and the maximum deviation reaches 5.57 %. At the same time, the trend of the experimental values is consistent with the simulated values and they are relatively stable with the variation of concentration, which indicates that the concentration has little effect on the CoV of the fluid distribution. This further confirms the accuracy and applicability of the model under laminar flow conditions.

4.2. Comparison of the mixing performance of different structures

4.2.1. Pressure drop

In fluid mechanics, Z_f is used to describe the relative pressure loss caused by static mixers, which is an essential index to evaluate the efficiency of different static mixers. It could also quantify the ratio of the pressure loss increased by the mixer to the pressure loss of the empty pipe [29,55,66].

$$Z_f = \frac{\Delta p}{\Delta p_{\text{empty pipe}}} \tag{16}$$

where Δp represents the pressure drop after the fluid passing through the static mixer, while $\Delta p_{\text{empty pipe}}$ is the pressure drop in the empty pipe under the same working conditions, which is determined by the Hagen-Poiseuille equation for laminar flow condition in a horizontal pipe.

$$\Delta p_{\text{empty pipe}} = \frac{64}{Re_g}\frac{\rho \bar{v}_z^{\,2}}{2}\frac{l_m}{D} \tag{17}$$

The Reynolds number of non-Newtonian fluids is defined by the generalized Reynolds Number (Re_g) given by Metzner [67]:

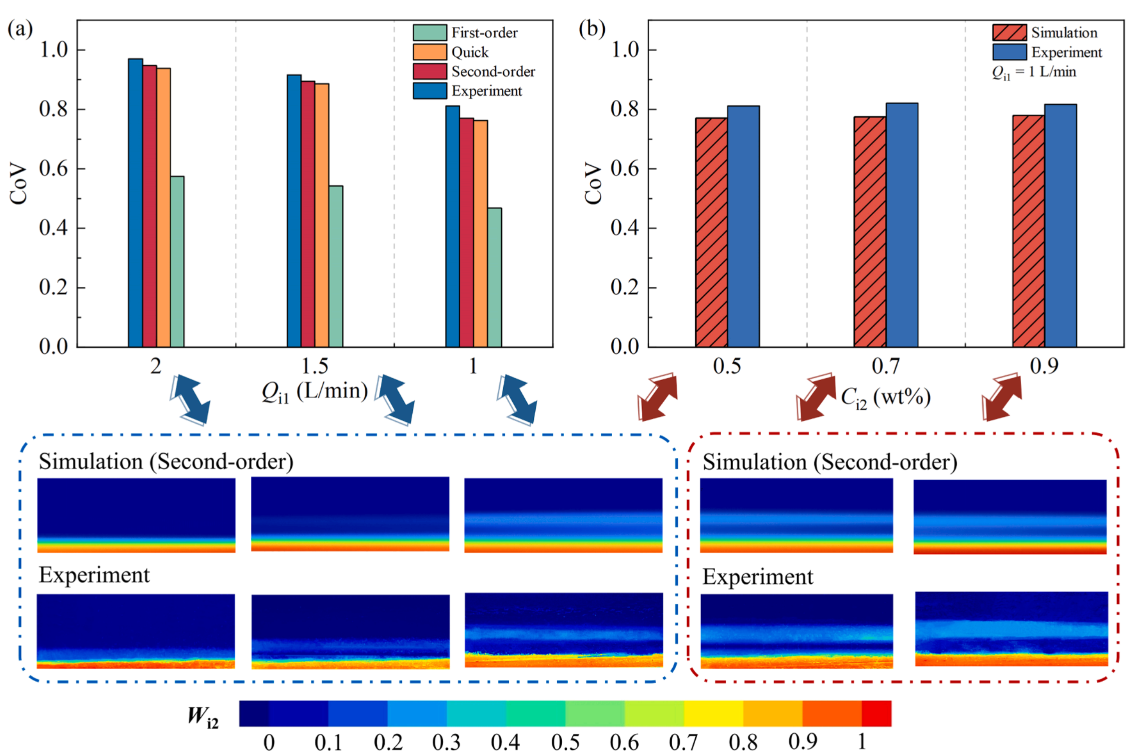

Fig. 6. Model validation for simulation and experimental data: (a) CoV of different discretization schemes and experiment at $Q_{i1} = 1$,1.5 and 2 L/min and (b) CoV for various C_{i2} at $Q_{i1} = 1$ L/min.

$$Re_g = \frac{\rho \bar{v}_z^{\,2-n} D^n}{8^{n-1} K'} \tag{18}$$

where K' is the consistency coefficient of the generalized non-Newtonian fluids.

For power-law fluids, Re_g is defined as follows:

$$Re_g = \frac{\rho \bar{v}_z^{\,2-n} D^n}{8^{n-1} \left(\frac{3n+1}{4n}\right)^n K} \tag{19}$$

The mixing performance of four different static mixers and empty pipe is evaluated by comparing the relationship between Δp as well as Z_f and Q_{i1} as presented in Fig. 7. It can be speculated that the Δp of all static mixers and empty pipe increases as Q_{i1} rises when Q_{i1} ranges from 0.25 to 2.5 L/min. Compared with the empty pipe, the increment of Δp in Kenics, Komax, S-Type and S-Type-b static mixers increases by 7.59 times, 8.10 times, 8.00 times and 6.25 times, respectively. The results indicate that Komax has a triangular flow channel formed by inclined slotted plates compared with other static mixers, which increases the complexity of fluid flow and the increment of Δp. The S-Type-b mixer removes two side baffles compared with the S-type mixer, which not only optimizes the flow path, but also reduces the local resistance of fluid flow.

It can be seen from the line chart in Fig. 7 that Z_f curves of Kenics and S-Type, as well as Komax and S-Type-b exhibit similar variation trend. Compared with Kenics, Komax and S-Type, the Z_f of S-Type-b decreases by 5.60–17.86 %, 10.60–21.58 % and 13.25–24.98 %, respectively. This could be attributed to the alternately arranged helical elements in Kenics, which induce the variation of local flow velocity at certain flow rates leading to the fluctuation of Z_f. Although the blade elements of S-Type may not generate intense rotation effect like the helical elements of Kenics, its symmetrical layout could cause similar fluid division and mixing dynamic characteristics thus producing a similar fluctuation trend of Z_f. The Z_f trend of Komax is similar to that of S-Type-b, because Komax forms a triangular flow channel. But S-Type-b also forms a similar flow channel through simplifying its internal structure, which reduces the flow resistance and produces a shear effect when the fluid passes through, thus keeping a relatively stable Z_f trend within the flow rates range. On the whole, S-Type-b shows outstanding performance in reducing Δp especially in the higher range of flow rates.

4.2.2. Mixing index

The *MI* is introduced to compare the distribution mixing performance of different static mixers. The parameter was utilized by Viktorov et al. [68] and Kaid et al. [69] to quantify the mixing degree in the mixer. The ideal mixing state means that different phases of fluids are completely and uniformly distributed at any point in the cross section of

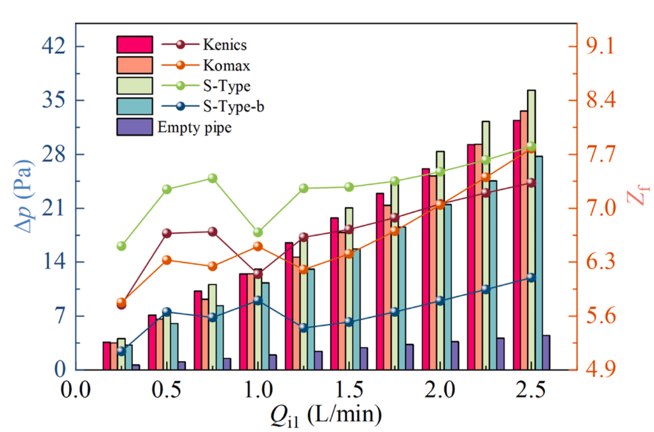

Fig. 7. Comparison of Δp and Z_f of different static mixers and empty pipe at different Q_{i1}.

the static mixer outlet. When the *MI* approaches 1, it manifests a higher mixing uniformity and almost reaches the ideal state.

$$\sigma_{max} = \sqrt{\bar{c} \cdot (1 - \bar{c})} \tag{20}$$

$$MI = 1 - \frac{\sigma}{\sigma_{max}} \tag{21}$$

where σ_{max} represents the maximum standard deviation of concentration, which is typically determined by the flow rates ratio of the inlet.

The *MI* change of four static mixers at different cross sections under various flow rates is illustrated in Fig. 8. The variation of *MI* for these mixers at different z/l sections presents specific regularity with the increase of flow rates. The *MI* of Kenics, Komax, S-Type and S-Type-b static mixers increases as z/l rises, which indicates that the mixing effect is enhanced with increasing mixing elements. Compared with $Q_{i1} = 0.25$ L/min, the *MI* of various static mixers at $z/l = 1$ under $Q_{i1} = 2.5$ L/min increases by 1.38 times, 2.19 times, 1.44 times and 2.03 times, respectively. This suggests that the increase of flow rates makes the shearing action of the fluids stronger when it passes through the mixing element thus improves the mixing uniformity. The growth curves of *MI* for four static mixers after $z/l = 5$ tend to stabilize. The mixing of the fluids in mixers is adequate, which makes the *MI* reach a relatively high value. If the number of mixing elements continues to increase, it also increases the Δp and reduce flow velocity, so the mixing quality would not be greatly improved. The *MI* of S-Type-b mixer at $z/l = 8$ under different flow rates is higher by 0.29–2.78 %, 1.36–4.02 % and 0.18–4.43 % than that of Kenics, Komax and S-Type mixers, respectively. This indicates that the S-Type-b static mixer possesses advantages of superior mixing effect and distributed performance. It can be inferred that the S-Type-b static mixer has superior distribution mixing performance.

The concentration distribution of four types of static mixers at various cross sections with $Q_{i1} = 0.25$ L/min and $Q_{i2} = 0.2$ L/min are evaluated as depicted in Fig. 9. The concentration distribution at the inlet cross section of $z = -0.06$ m forms a circular fluid mass. The initial secondary flow mass is divided into two fluid masses after being cut by the first mixing element. In the subsequent mixing process of the five elements, two fluid masses are continuously stretched, cut and folded, forming a striation-like distribution and the number of striations is increasing. The reason is that the fluid distribution in the early stage of the mixing section is primarily influenced by stretching and cutting and it would be cut by the mixing elements when the stretching reaches a certain extent. The formation and evolution of striations are due to the inherent fluid dynamics characteristics in the design of static mixers, in which fluids are forced to pass through narrow channels and complicated flow paths increasing the contact area and mixing quality between fluids. An increase in the number of striations suggests that the mixing is enhanced and the interfacial area between fluids is also correspondingly increased, which contributes to strengthen the diffusion and mixing between molecules. As the fluid passes through the last three elements, the fluid distribution becomes more uniform and the striation characteristics disappear, which signifies the efficient completion of the mixing process and the injection pipe fluid is fully dispersed and homogenized. It is worth paying attention to that the yellow striations of S-Type-b almost disappear after $z = 0.24$ m and S-Type-b mixer enters the uniform mixing state faster than other static mixers, demonstrating that the novel design optimizes the stretching, cutting and folding processes of the fluid and strengthens the mixing effect.

4.2.3. Power consumption

The index to measure the energy efficiency of static mixers was proposed to describe the relationship between required power and mixing efficiency, which is defined as the power consumption (*Pc*) per unit volume of mixer when the target mixing quality is achieved [41,70,71]:

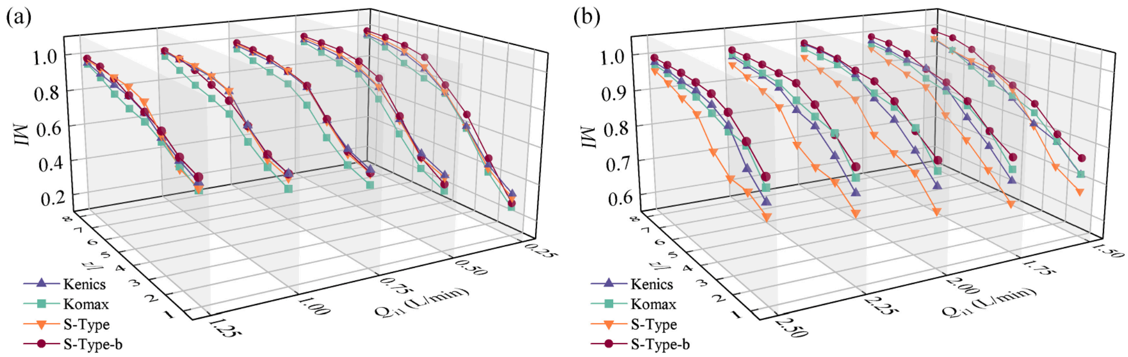

Fig. 8. The *MI* of different static mixers at different cross sections under various flow rates: (a) Q_{i1} = 0.25–1.25 L/min and (b) Q_{i1} = 1.50–2.50 L/min.

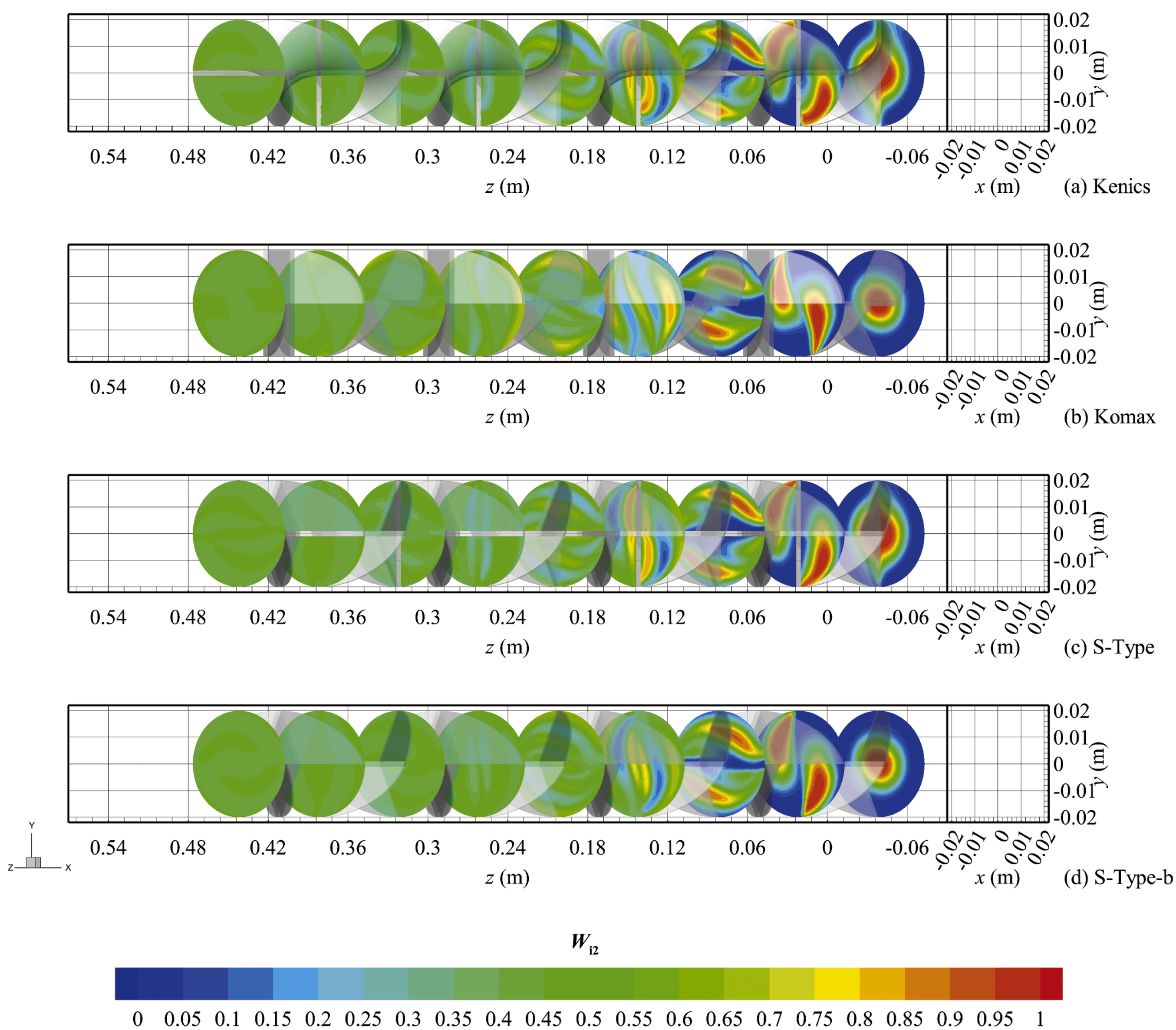

Fig. 9. W_{i2} for static mixers at different cross sections under Q_{i1} = 0.25 L/min and Q_{i2} = 0.2 L/min: (a) Kenics, (b) Komax, (c) S-Type and (d) S-Type-b.

$$Pc = \frac{\Delta p_m \times Q}{V_m} \tag{22}$$

where Δp_m and V_m are the pressure drop and effective volume of the mixer when *MI* reaches 0.95, respectively.

It can be observed from Fig. 10 that the power consumption of all mixers nonlinearly increases with the rise of flow rate, but the range of increase significantly varies because of different designs. The power consumption of S-Type-b mixer is 1.70 – 1.90 times, 1.75 – 1.84 times and 1.95 – 2.06 times lower than that of Kenics, Komax and S-Type mixers, respectively. This is because S-Type-b not only reduces the energy loss, but also optimizes the flow field distribution and enhances the

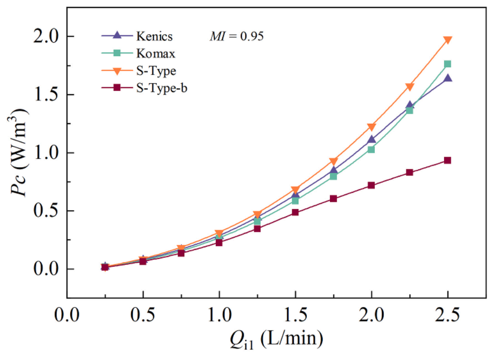

Fig. 10. Comparison of Power Consumption of different static mixers when *MI* is 0.95 at different flow rate.

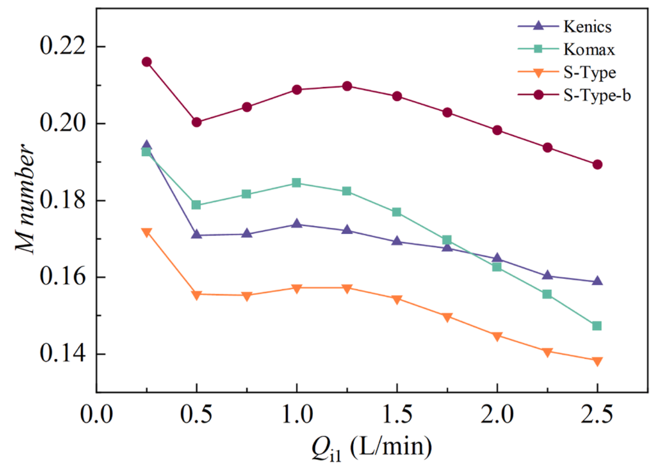

Fig. 11. The *M number* of four static mixers at different flow rates.

shear rate by removing the side baffle, which makes it achieve efficient mixing at low power consumption. The geometric structure of S-Type is not enough to balance the relationship between power input and mixing efficiency, which causes a higher power consumption. The design of Kenics and Komax mixers failed to effectively reduce the resistance at high flow rate, resulting in a rapid increase in power consumption.

4.2.4. *M number*

The quantitative index *M number* was proposed and derived by Medina et al. [72], which provided new insights for the design and optimization of static mixers. The *M number* essentially offers an effective method to evaluate the overall performance of static mixers, taking into account two key factors of mixing efficiency and energy consumption. An ideal static mixer could not only provide superior mixing effect but also have low energy consumption. If the value of the *M number* is closer to 1, it demonstrates that the mixer could achieve excellent mixing effect with less energy consumption. When the value of the *M number* is low, the mixer either has poor mixing effect or high energy consumption, or both factors may exist.

$$z_1 = \sum_{i=1}^{N} \frac{a_{1i}}{A}(c_{1i}) \tag{23}$$

$$\zeta_M = \frac{\sum_{i=1}^{N} \frac{a_{1i}}{A}\left[c_{1i}\ln c_{1i} + (1-c_{1i})\ln(1-c_{1i})\right]}{z_1 \ln z_1 + (1-z_1)\ln(1-z_1)} \tag{24}$$

where A represents the total cross-sectional area of the static mixer and a_i denotes the area of the ith cell in the cross sections. z_1 is the weighted average of concentration, where the weight is determined by the grid area. ζ_M represents mixing efficiency parameter based on information entropy calculation. The uncertainty or randomness of mixing is quantified by the entropy of concentration distribution as shown in Eq. (25).

$$M\ Number = \frac{\dfrac{\left(\zeta_{M,out} - \zeta_{M,out,empty\ pipe}\right)}{\left(1 - \zeta_{M,out,empty\ pipe}\right)}}{\dfrac{P_{in}}{P_{in} - P_{out,empty\ pipe} + P_{out}}} \tag{25}$$

where $\zeta_{M,out}$ and $\zeta_{M,out,empty\ pipe}$ are the mixing efficiency at the outlet of the static mixer and empty pipe, respectively. P_{in} and P_{out} refer to the pressure at the inlet and outlet of the static mixer, respectively. $P_{out,empty\ pipe}$ stands for the outlet pressure of the empty pipe.

The variation of *M number* of four different static mixers with Q_{i1} is illustrated in Fig. 11. When Q_{i1} increases from 0.25 L/min to 0.5 L/min, the *M number* of Kenics, Komax, S-Type and S-Type-b decreases by 11.96 %, 7.14 %, 8.53 % and 7.24 %, respectively. This may be because the fluid shear force in the mixer increases with the increase of flow rates, but the mixing efficiency does not significantly improve. Meanwhile, the pressure loss of the system rises leading to the decline of the overall mixing performance. Subsequently, the *M number* slowly rises as the flow rates increase from 0.5 L/min to 1 L/min because the mixers begin to adapt to higher flow rates and the increase rate of pressure loss slows down at the same time. It is worth noting that the *M number* of the S-Type-b static mixer increases when the Q_{i1} rises from 0.5 L/min to 1.25 L/min, which suggests that the S-Type-b static mixer performs better in this range and exhibits a wider flow rates range than the other three static mixers. It can be observed that the *M number* continues to decrease again as the flow rates further increase to 2.5 L/min. This implies that although the mixing efficiency of mixers is improved at higher flow rates, it cannot offset the increase in energy consumption caused by pressure loss. The *M number* of S-Type-b static mixer is higher by 10.13–18.29 %, 10.82–22.26 %, and 20.39–27.35 % than that of Kenics, Komax, and S-Type, respectively. As a result, the S-Type-b static mixer could achieve an optimal balance between mixing quality and energy consumption through its innovative design. Its optimized structure maximizes shear forces for efficient mixing and minimizes unnecessary pressure loss.

4.2.5. *Extensional efficiency*

The dispersive mixing process in static mixers is influenced by fluid dynamics, which is determined by its geometric design. The existence of mixing elements introduces more complex flow types that are essential to enhance mixing especially in high viscosity laminar flow, which are crucial for the effective dispersion and homogenization of fluids in static mixers [73,74].

$$\alpha = \frac{|\gamma|}{|\gamma| + |\omega|} \tag{26}$$

$$|\gamma| = \frac{1}{2}\left[\nabla \boldsymbol{v} + (\nabla \boldsymbol{v})^{\mathrm{T}}\right] \tag{27}$$

$$|\omega| = \frac{1}{2}\left[\nabla \boldsymbol{v} - (\nabla \boldsymbol{v})^{\mathrm{T}}\right] \tag{28}$$

where γ is the deformation rate tensor, ω is vorticity tensor and $\nabla \boldsymbol{v}$ is the velocity gradient. The α quantifies the deformation of fluids in mixer elements. For $\alpha = 1$, the deformation only involves shear or extension, which optimizes mixing efficiency and minimizes energy consumption. When $\alpha = 0.5$, rotation and shear deformation coexist causing appropriate mixing and higher energy consumption. On the contrary, $\alpha = 0$ signifies pure rotation motion and the mixing efficiency is the lowest.

The average α of four static mixers at various cross sections under different operation conditions is illustrated in Fig. 12. It can be seen that Komax and S-Type-b exhibit obvious peaks at the initial position of each mixing element, while the fluctuation amplitude is small in the middle position of elements. It can be concluded that the dispersive mixing effect at the transition of elements is more prominent. However, there is mainly distribution mixing in the middle part of the element. In contrast, the peaks of Kenics and S-Type are not significant at the starting position of different elements (except for the first element). This phenomenon could be attributed to the unique design features of Komax and S-Type-b static mixers. The design of Komax promotes more fluid interaction within a small space and effectively enhance the shear and stretching effects of fluids further improving the mixing efficiency and uniformity. The improved design of S-Type-b provides a more effective dispersion mixing effect by increasing the free fluidity of the fluid. But the blade design of Kenics and S-Type is relatively weak in promoting the shear and stretching of fluid mass. When the flow rates increase, it could be observed that the comprehensive average α of all cross sections is less affected by the flow rates and fluctuates around a fixed value, which has also been confirmed in the research of Chakleh et al. [41].

The probability density distribution of the average α for four static mixers under different various operation conditions is presented in Fig. 13, which can comprehensively compare the α of different static mixers. It can be evidently observed that the average extension efficiency of all static mixers is basically within the range of 0.46 to 0.62. The four static mixers are mainly distributed around 0.52 under four different working conditions suggesting that the flow in static mixers is primarily characterized by simple shear flow. Based on the percentage of probability density distribution of the average extension efficiency for Kenics, Komax, S-Type and S-Type-b in the range of 0.56 to 0.62, Komax performs the best in this range with a distribution percentage ranging from 14.11 % to 28.51 %, demonstrating its superior mixing efficiency. In comparison, the performance of Kenics is relatively poor only with a distribution percentage ranging from 5.93 % to 6.20 %. The S-Type is almost unable to achieve this extension efficiency and shows the lowest distribution percentage of 0 to 3.71 %. While the S-Type-b also performs better with a distribution percentage of 12.45 % to 22.31 %. In summary, the improvement of the geometric model of S-Type-b static mixer is very effective and the α of S-Type-b static mixer is significantly improved.

The contour plots of α for four static mixers at $z = 0.24$–0.27 m under $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min are depicted in Fig. 14. It can be seen from Fig. 14(a) that the higher α area of Kenics at $z = 0.24$ m is primarily distributed in the middle area between the pipe wall and the element wall. However, the region with low extension efficiency is mainly concentrated near the wall surface of the element, because the rotational speed of the fluid micelle is high, which leads to a large rotational tension. However, the area with low α is primarily concentrated near the wall surface of elements, because the rotation rate of the fluid mass is large resulting in a larger rotation tensor. The area of high α presents a symmetrical distribution centered on the blades in the range of 0.246–0.27 m. This distribution phenomenon is due to the fact that adjacent blades are placed vertically and rotate in opposite directions at the transition of elements, which forces the fluid to alter the direction of radial secondary flow thus enhancing the renewal of interfacial and causing great deformation of the fluid mass. The S-Type mixer in Fig. 14 (c) exhibits a similar trend compared with Kenics. However, the complex blade design of the S-Type mixer increases the fluid disturbance in the flow process. Although there exists always a lower α area at the blade joint between $z = 0.246$ m and $z = 0.264$ m, this design reduces flow dead zones by enhancing the radial and axial mixing efficiency thus promoting the overall α. The Komax static mixer has high α at $z = 0.246$ m and $z = 0.252$ m as shown in Fig. 14(b). The distribution of α becomes more uniform but the average α decreases at $z = 0.264$ m and $z = 0.27$ m, which is because of the reduction of flow disturbance caused by mixing elements. The α distribution of the S-Type-b mixer at different positions is illustrated in Fig. 14(d). The α distribution of S-Type-b mixer is similar to that of Komax at $z = 0.24$ m, although the high α area is smaller than that of Komax and S-Type at $z = 0.246$ m, the overall α distribution is very uniform. The α distribution of S-Type-b mixer is similar to that of S-Type between $z = 0.252$ m and 0.27 m, but S-Type-b has a larger area of high α. This indicates that the removal of side baffles enhances the shear and stretch deformation of fluid mass, which further promotes dispersive mixing ability.

4.2.6. Chaotic characteristic

The Lagrange particle tracking method could be used to effectively study complex streamlines in the flow field. A certain number of tracer particles are released at the inlet cross section of the static mixer and each tracer particle is given a vector $\boldsymbol{I}$. This vector is stretched and deformed with the motion of the tracer particles when the fluid flows through the static mixer and the stretching vector of each particle can be calculated by the deformation of this vector. According to the mixing theory proposed by Ottino [75,76], the evolution of this vector can be calculated by the following formulas:

$$\frac{dz}{dt} = v(z) \tag{29}$$

$$\frac{d(\boldsymbol{I})}{dt} = (\nabla v)^{\mathrm{T}} \cdot \boldsymbol{I}, \boldsymbol{I}_{t=0} = \boldsymbol{I}_0 \tag{30}$$

where $\boldsymbol{I}$ denotes the stretch vector of particles and $\boldsymbol{I}_0$ signifies the initial stretch vector of particles.

The calculation of the Lyapunov exponent is considered as one of the most effective tools to characterize chaotic systems which are often unpredictable [77]. A positive Lyapunov exponent means that the system is very sensitive to initial conditions and any slight change would exponentially increase with time, which would lead to the unpredictability of system behaviors. Conversely, if the Lyapunov exponent decreases or even becomes a negative value, the trajectory in the system tends to converge rather than diverge. In the meantime, the system becomes more predictable and has a lower sensitivity to initial conditions.

The cumulative stretching vector of stretching degree for particles in a specific cross section is defined as:

$$\lambda = \frac{|\boldsymbol{I}|}{|\boldsymbol{I}_0|} \tag{31}$$

The average Lyapunov exponents of the KMX and SMX static mixers was calculated and compared by Heniche et al. [74]. LE_{ave} could be computed by the following formulas:

$$LE_{\mathrm{ave}} = \lim_{\mathrm{k}\to\infty}\left(\frac{1}{T_{z/l}}\frac{1}{\mathrm{k}}\sum_{\mathrm{k}=0}^{\mathrm{k}-1}\ln\lambda_{\mathrm{k}}\right) \tag{32}$$

$$\overline{LE}_{\mathrm{ave}} = \frac{l}{z}\sum_{z/l=0}^{z/l} LE_{\mathrm{ave}} \tag{33}$$

where λ_{k} represents the stretch rate of the kth particle, and $T_{z/l}$ denotes the average time for all particles to pass through the z/l element. The whole expression involves taking the logarithm of the local Lyapunov exponents for all k particles, averaging it, dividing the time for all k particles to pass through the $T_{z/l}$ element, and subsequently take the limit to calculate the final value. A decrease in the value of LE_{ave} typically means that the sensitivity of systems to initial conditions decreases with time. The LE_{ave} would converge to $\ln(\lambda_{\mathrm{k}})/T_{z/l}$ after the fluid passes through the z/l element.

Fig. 15 depicts the variation trend of $\overline{LE}_{\mathrm{ave}}$ for four static mixers at all cross sections under a variety of Q_{i1}. As Q_{i1} increases, the $\overline{LE}_{\mathrm{ave}}$ for all mixers is significantly promoted indicating that the sensitivity of the system to initial conditions and the chaotic behavior of systems are strengthened with the increase of flow rates. Specifically, the $\overline{LE}_{\mathrm{ave}}$ of Kenics static mixer is low at low flow rates suggesting that the system has a lower sensitivity to initial conditions. But when the Q_{i1} reaches 2 L/min, the $\overline{LE}_{\mathrm{ave}}$ is close to 10 and 1.01 – 1.18 times that of other static

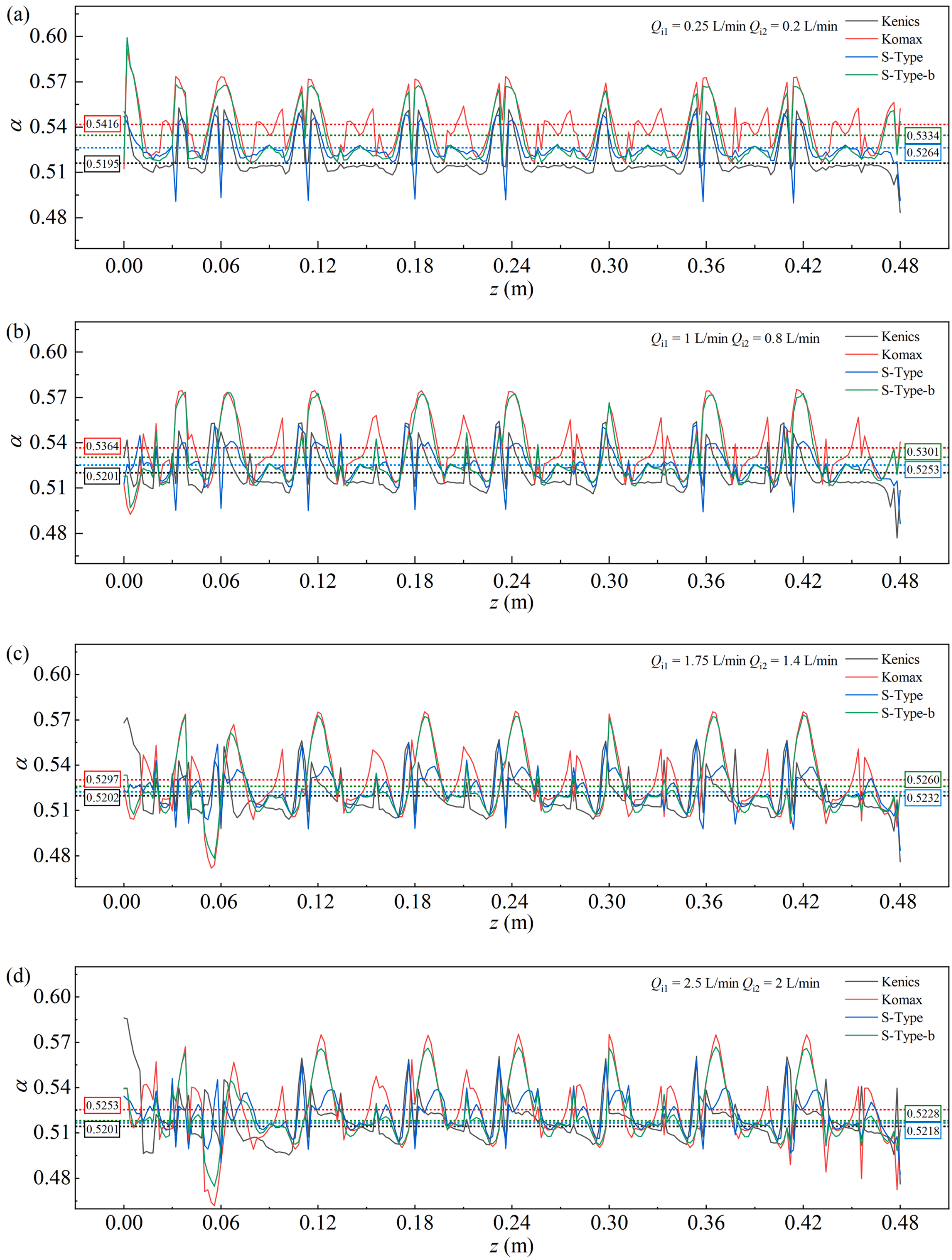

Fig. 12. The average α of various static mixers under four different flow rates: (a) $Q_{i1} = 0.25$ L/min and $Q_{i2} = 0.2$ L/min, (b) $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min, (c) $Q_{i1} = 1.75$ L/min and $Q_{i2} = 1.4$ L/min and (d) $Q_{i1} = 2.5$ L/min and $Q_{i2} = 2$ L/min.

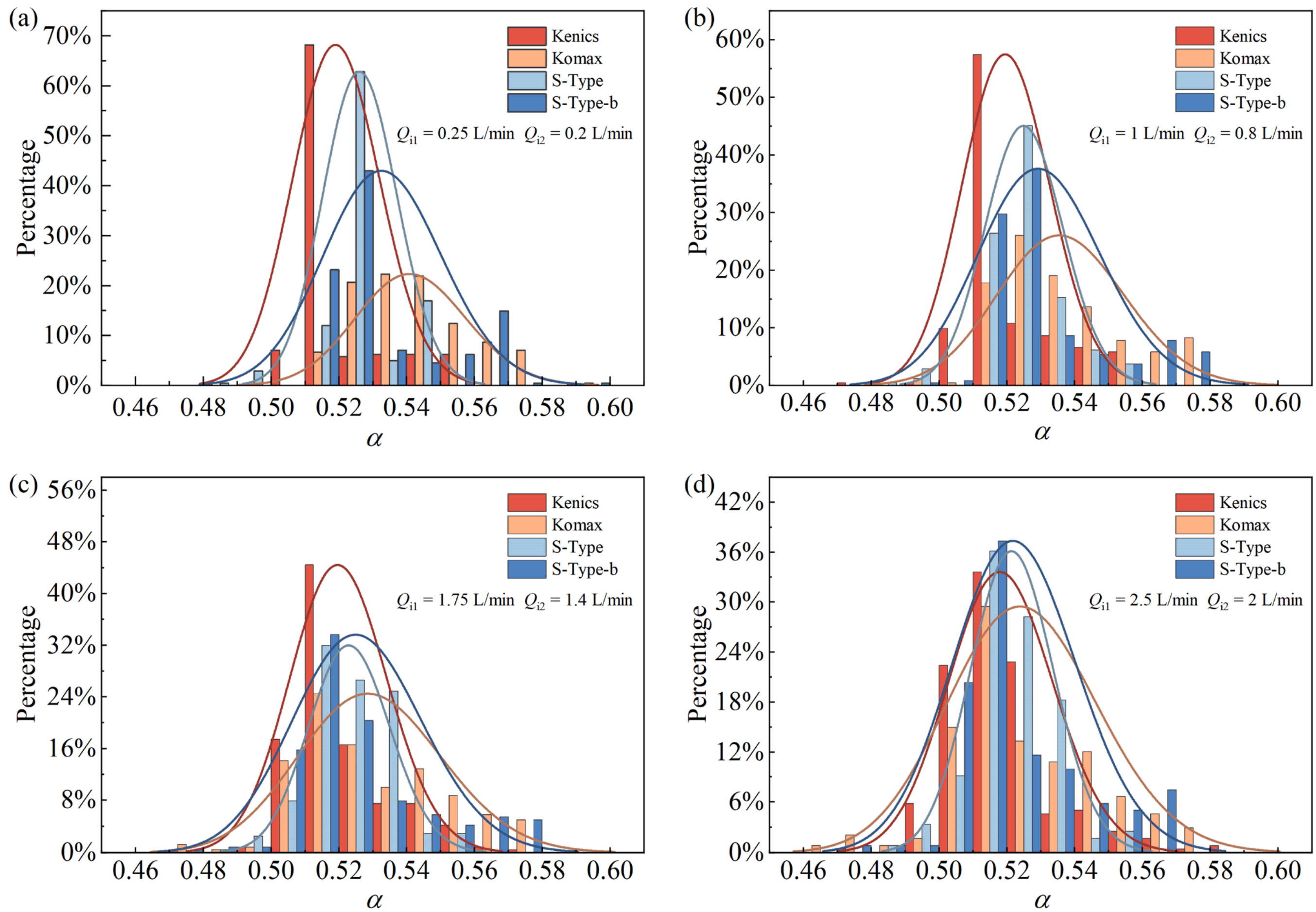

Fig. 13. The probability density distribution of α of various static mixers at different flow rates: (a) $Q_{i1} = 0.25$ L/min and $Q_{i2} = 0.2$ L/min, (b) $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min, (c) $Q_{i1} = 1.75$ L/min and $Q_{i2} = 1.4$ L/min and (d) $Q_{i1} = 2.5$ L/min and $Q_{i2} = 2$ L/min.

mixers, which demonstrates that its sensitivity to initial conditions is significantly enhanced. The $\overline{LE}_{ave}$ of Komax static mixer under most flow rates is lower than that of other static mixers indicating that it has lower sensitivity to initial conditions. The $\overline{LE}_{ave}$ of the S-Type static mixer at $Q_{i1} = 0.25$–1.25 L/min is 1.01 – 1.21 times than that of other static mixers, which shows higher sensitivity to initial conditions and the chaotic behaviors are strengthened, but its $\overline{LE}_{ave}$ is lower than other static mixers after $Q_{i1} = 1.25$ L/min. The S-Type-b static mixer is shown to display high across all flow rates, indicating it has a sustained chaotic behavior. The $\overline{LE}_{ave}$ of the S-Type-b mixer under most flow rates is high, which manifests continuous chaotic behaviors.

Nevertheless, LE_{ave} cannot accurately describe the overall chaotic behaviors because it may be relatively high in local areas. Therefore, LE_{var} is introduced as an additional index for evaluation.

$$LE_{var} = \frac{\sum_{k=0}^{k-1}(\lambda_k - LE_{ave})^2}{k} \tag{34}$$

where LE_{var} is helpful to assess the spatial variability of mixing efficiency thus providing a more comprehensive understanding of overall chaotic behavior of the system.

The variation trends of LE_{ave} for four static mixers under three different flow rates at different z/l are shown in Fig. 16(a). The increase rate of LE_{ave} also increases as flow rates rise. The LE_{ave} is enhanced with the increase of axial position indicating the improvement of mixing effect at $Q_{i1} = 0.5$ L/min, $Q_{i2} = 0.4$ L/min and $Q_{i1} = 1.25$ L/min, $Q_{i2} = 1$ L/min. Nevertheless, the LE_{ave} reaches the peak at $z/l = 7$ then declines slightly under the condition of $Q_{i1} = 2$ L/min and $Q_{i2} = 1.6$ L/min, which may reflect that periodic flow characteristics of the system and further cause the weakening of chaotic behaviors. Fig. 16(b) depicts the variation of LE_{var} for four static mixers with z/l under various flow rates. It is observed that all static mixers exhibit low LE_{var} at low flow rates, which suggests that fluid flow is relatively stable and has small fluctuation. The LE_{var} of all static mixers gradually increases with the rise of flow rates revealing that higher flow rates may bring more flow instability and the variation of mixing behavior. Although the Komax static mixer is smaller than other static mixers at $Q_{i1} = 0.5$ L/min, $Q_{i2} = 0.4$ L/min and $Q_{i1} = 1.25$ L/min, $Q_{i2} = 1$ L/min combined with Fig. 16(a) and (b), it exhibits lower flow fluctuation and LE_{var}. This demonstrates that higher chaotic behavior is not produced under these conditions, which might not be ideal for application scenarios requiring efficient mixing. In contrast, Kenics and S-Type static mixers show high LE_{ave}, manifesting strong chaotic behavior and efficient mixing effect, but they also show higher flow instability and fluctuation, which may cause problems in some application needing highly stable operation. The S-Type-b static mixer has higher LE_{ave} and lower LE_{var}, which shows that the mixing effect of S-Type-b mixer at different axial positions is remarkable and the

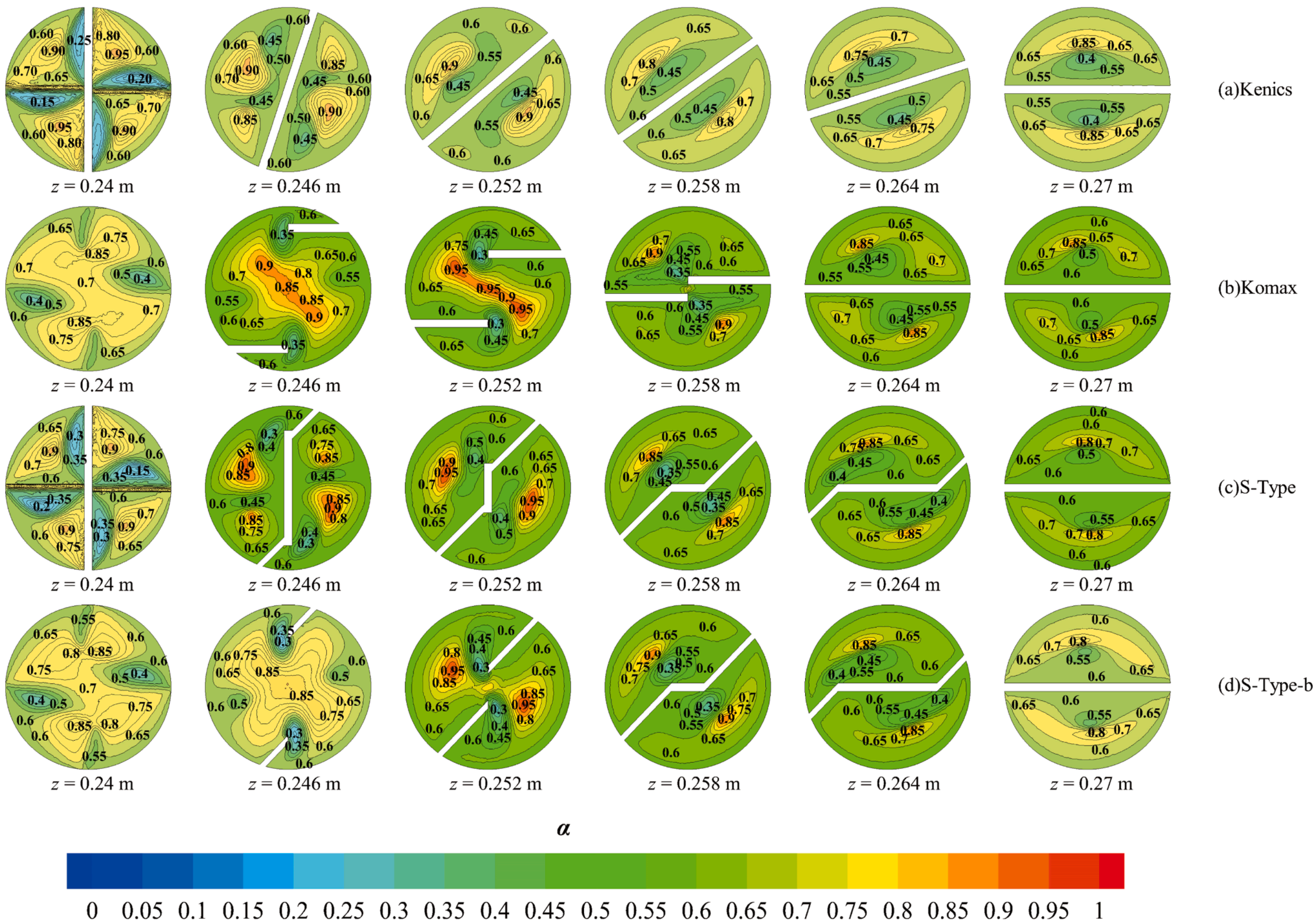

Fig. 14. The α distribution of four static mixers under $Q_{i1} = 1$ L/min and $Q_{i2} = 0.8$ L/min at $z = 0.24 - 0.27$ m: (a) Kenics, (b) Komax, (c) S-Type and (d) S-Type-b.

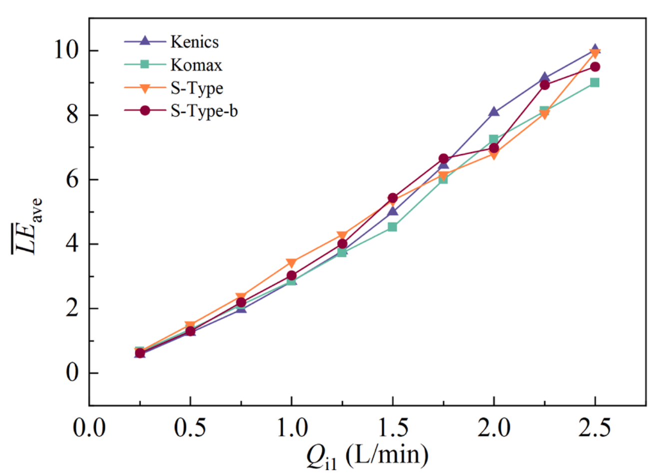

Fig. 15. The variation trend in $\overline{LE}_{ave}$ for four types of static mixers under different Q_{i1}.

sensitivity to initial conditions is high. It can introduce more chaos and complex flow structures into fluid dynamics and the fluctuation of mixing behavior is small as well as the flow is more stable. This balance between effective mixing and flow stability is advantageous in various practical applications.

4.3. Multi-objective optimization

4.3.1. Design variables and objective functions

The previous comparative studies have shown that the S-Type-b static mixer had superior mixing efficiency and chaotic characteristics. The Design Expert 13.0 is used for modeling in this study and five critical design variables and their ranges are selected from the perspectives of structural parameters of static mixers, operating conditions and fluid properties: A_r ranging from 1 to 2, flow ratio (Q_r) ranging from 0.4 to 1.5, length of inlet 2 (l_2) ranging from 0.15 m to 0.5 m, and the concentration of the main pipe fluid (C_{i1}) and concentration of the injection pipe fluid (C_{i2}) ranging from 0.5 wt% to 0.9 wt% as listed in Table 4.

The Optimal Design is an experimental design strategy in RSM, which is suitable for asymmetric variable range, irregular design space and high-dimensional problems. Compared with the Central Composite Design (CCD), the risk of the axis point beyond the experimental range could be effectively avoided through the Optimal Design [78–80]. Fig. 17 shows the procedure of designing the experimental scheme. The Optimal Design methodology is utilized to design 42 experimental schemes as shown in Table S1, aiming at optimize two key indexes (namely *M number* and $\overline{LE}_{ave}$) and further enhance the mixing efficiency of the S-Type-b static mixer.

4.3.2. Modeling with RSM

In ANOVA, F-value and p-value are crucial indexes to measure the significance of each variable in the model. A higher F-value typically indicates a greater reliability of variables, while a p-value is <0.05 denotes that the variable is statistically significant. Conversely, a p-value is greater than 0.1 suggests that the effect of variables is not significant.

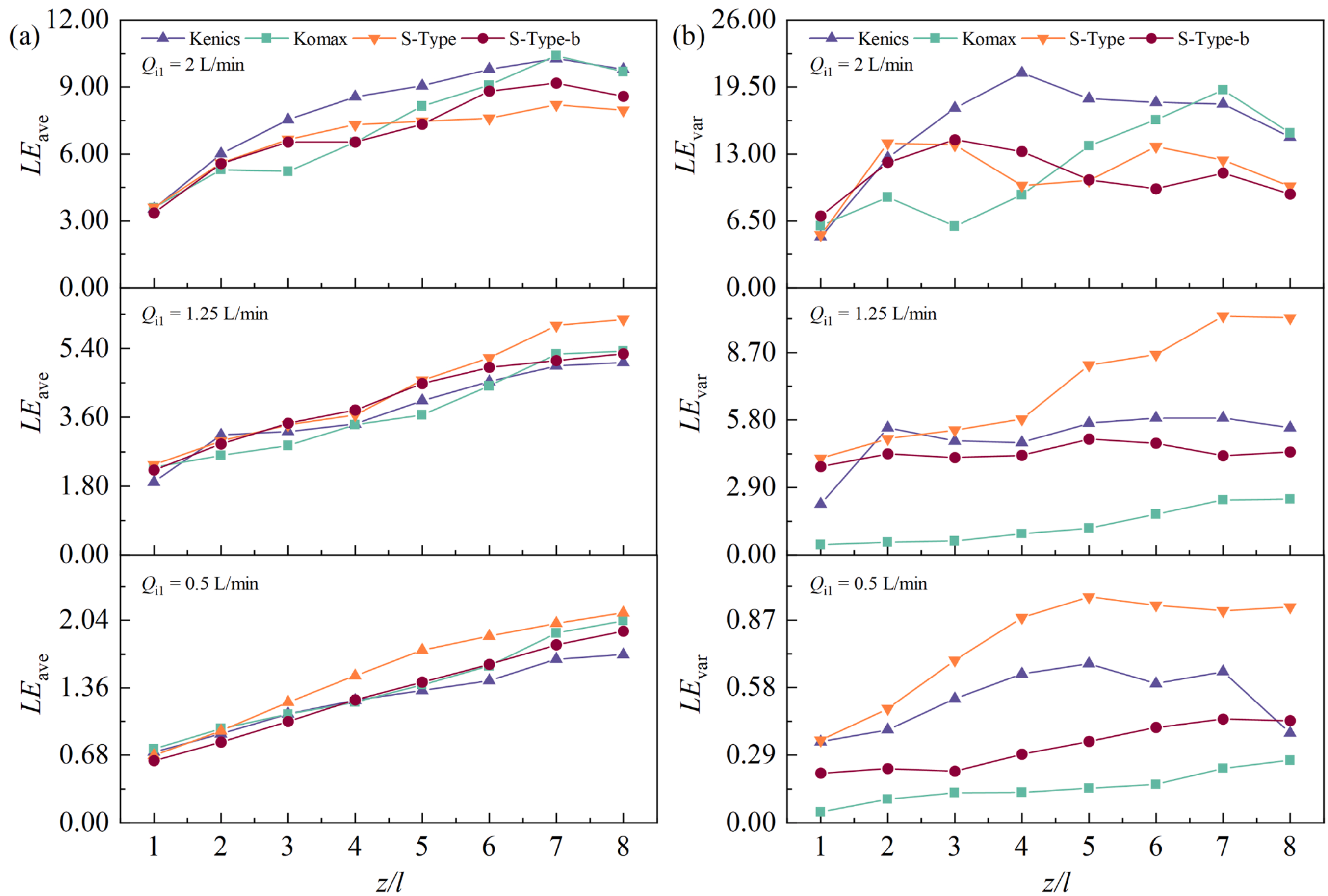

Fig. 16. The variation trends of four static mixers at different z/l positions under three flow rates: (a) LE_{ave} and (b) LE_{var}.

Table 4
Limits of design variables.

No.	Design variables	Lower limit	Upper limit
1	Aspect ratio, A_r	1	2
2	Flow ratio, Q_r	0.4	1.5
3	Length of inlet 2, l_2 (m)	0.15	0.5
4	Concentration of the main pipe fluid, C_{i1} (wt%)	0.5	0.9
5	Concentration of the injection pipe fluid, C_{i2} (wt %)	0.5	0.9

According to the data in Table S2, the single factor analysis of *M Number* demonstrates that the variable A_r has the most significant impact on the model followed by Q_r and l_2, and C_{i2} also shows statistical significance. In terms of interaction, the interaction between A_r and Q_r, A_r and C_{i2}, as well as Q_r and l_2 all reached a significant level, while other interaction terms do not exhibit significant influence. The A_r^2 and Q_r^2 show significant effects in the quadratic analysis, but the l_2^2, C_{i1}^2 and C_{i2}^2 do not show significant effects. Comparing the single factor analysis results of the $\overline{LE}_{ave}$ in Table S3, it is found that A_r remains the most significant influence factor followed by Q_r and l_2, while C_{i1} and C_{i2} also show significant effects. Only the interaction between A_r and Q_r is significant among the interaction, while other interaction is not significant. For quadratic analysis, only A_r^2 is significant, while Q_r^2, l_2^2, C_{i1}^2, and C_{i2}^2 are not significant.

The regression analysis is used to estimate the mixing efficiency of the S-Type-b static mixer. The regression equations for the *M number* and $\overline{LE}_{ave}$ are shown in Eqs. (35) and (36), and their application ranges are presented in Table 4.

$$\begin{aligned} M\ number = {} & -0.411336 + 0.575234 \times A_r - 0.0725424 \times Q_r - 0.0444734 \times l_2 + 0.134934 \times C_{i1} \\ & + 0.24799 \times C_{i2} + 0.0475152 \times A_r \times Q_r + 0.0317434 \times A_r \times l_2 - 0.0353097 \times A_r \times C_{i1} \\ & - 0.104764 \times A_r \times C_{i2} - 0.0457759 \times Q_r \times l_2 - 0.0228888 \times Q_r \times C_{i1} - 0.0228267 \times Q_r \\ & \times C_{i2} - 0.0872218 \times l_2 \times C_{i1} - 0.0304335 \times l_2 \times C_{i2} - 0.0121646 \times C_{i1} \times C_{i2} - 0.12053 \\ & \times A_r^2 + 0.0314968 \times Q_r^2 + 0.147975 \times l_2^2 - 0.0129676 \times C_{i1}^2 - 0.0504039 \times C_{i2}^2 \end{aligned} \tag{35}$$

$$\begin{aligned} \overline{LE}_{ave} = {} & 11.8809 - 11.2907 \times A_r + 7.13679 \times Q_r + 4.05466 \times l_2 - 2.207 \times C_{i1} - 3.90855 \times C_{i2} \\ & - 3.72398 \times A_r \times Q_r - 1.81932 \times A_r \times l_2 - 0.353503 \times A_r \times C_{i1} - 0.315935 \times A_r \times C_{i2} \\ & + 0.907279 \times Q_r \times l_2 - 0.402899 \times Q_r \times C_{i1} - 0.213215 \times Q_r \times C_{i2} - 5.13229 \times l_2 \times C_{i1} \\ & + 0.168711 \times l_2 \times C_{i2} + 0.22844 \times C_{i1} \times C_{i2} + 3.94749 \times A_r^2 + 0.797488 \times Q_r^2 + 4.73181 \\ & \times l_2^2 + 2.42088 \times C_{i1}^2 + 1.89974 \times C_{i2}^2 \end{aligned} \tag{36}$$

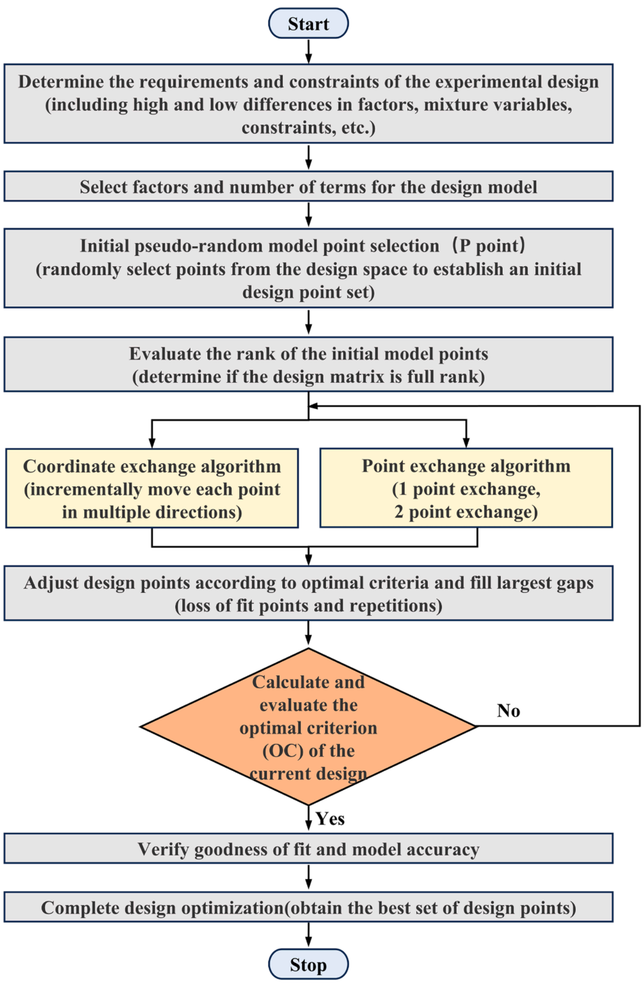

Fig. 17. Experimental design process of optimal design.

Fig. 18 presents the comparison between actual and predicted values for two datasets and the prediction accuracy of the model is intuitively demonstrated by scatter plots and fitting lines. The colored dots represent the actual observed values, and the red lines are the results predicted by the model. The R^2 value of *M number* in Fig. 18(a) is as high as 0.9944, which reveals that the model could capture and predict the data variation with high precision. The data points are closely distributed around the predictive line, illustrating the high degree of consistency and remarkable explanatory ability of the model in this dataset. The R^2 value of the $\overline{LE}_{\mathrm{ave}}$ in Fig. 18(b) is 0.9869, which is slightly lower than that in Fig. 18(a), but it still shows that the model has strong fitting capability and high predictive accuracy for this dataset.

4.3.3. Response surface analysis

The contour map and response surface plots of the interaction term in *M number* is illustrated in Fig. 19, in which the A_r has the most significant influence on *M number*. The *M number* gradually rises as A_r increases, which is primarily due to simplified fluid path, further reducing the pressure loss. Nevertheless, the pressure loss within the static mixer also rises when C_{i1} and C_{i2} increase resulting in the decrease of *M number*. This is because higher concentration increases the viscosity and density of fluids, which further enhances the flow resistance and Δp. Additionally, *M number* initially decreases then increases with the rise of Q_r and l_2. In the initial increase stage of the Q_r, the Δp increases with the increased flow resistance, resulting in the decline of *M number*. However, the fluid mixing effect is strengthened with the further increase of the Q_r, making the flow more stable, the local pressure loss reduced and *M number* increased. Likewise, *M number* is higher because of the smaller flow resistance when is short. The flow path of fluids is prolonged with the increase of and the initial Δp is increased, causing the decline of *M number*. However, the further increase of the l_2 provides more time and space for complete mixing, which leads to the rise in *M number* again.

Fig. 20 shows the contour map and response surface plot of the interaction terms in the $\overline{LE}_{\mathrm{ave}}$, where the A_r has the same significant influence on $\overline{LE}_{\mathrm{ave}}$. The $\overline{LE}_{\mathrm{ave}}$ shows an upward trend as A_r decreases, indicating that the smaller A_r makes the fluid path more complicated thus enhancing the chaos of the system and its sensitivity to initial conditions. On the contrary, the increase in C_{i1} and C_{i2} leads to the decline of the $\overline{LE}_{\mathrm{ave}}$, which may be because the higher concentration makes the system tend to a more stable state thus reducing the chaotic characteristics. Furthermore, the $\overline{LE}_{\mathrm{ave}}$ also shows an upward trend with the increase of the Q_r and l_2. The increase of the Q_r results in the complexity and unpredictability of fluid path, which would affect the dynamic stability of the whole system. A longer l_2 allows flow have more development and disturbance, especially in high-speed flow and complex geometry, which enhances flow instability and chaotic behaviors.

4.3.4. Optimization of designing parameters

Larger population can usually explore the solution space more quickly and maintain the diversity of the solution space, while the increase of iteration times provides more time for the algorithm to select, cross and mutate thus promoting the possibility of finding the Pareto-optimal solutions. Nevertheless, too large population and iteration times may increase the consumption of computation resources and also reduce the diversity by increasing the similarity among individuals [81].

Fig. 21 depicts the optimal solution distribution of different iteration times when the population sizes are 200, 300 and 500. It can be

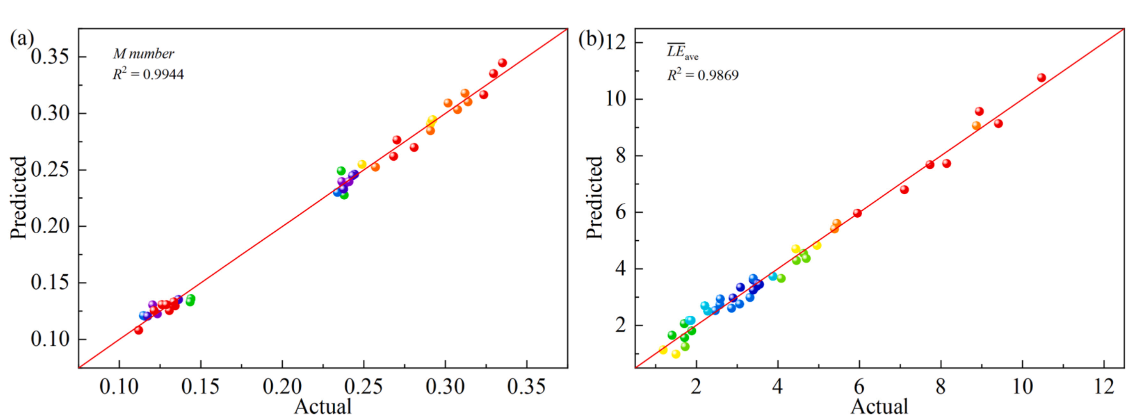

Fig. 18. Comparison between actual and predicted values for two datasets: (a) *M number* and (b) $\overline{LE}_{\mathrm{ave}}$.

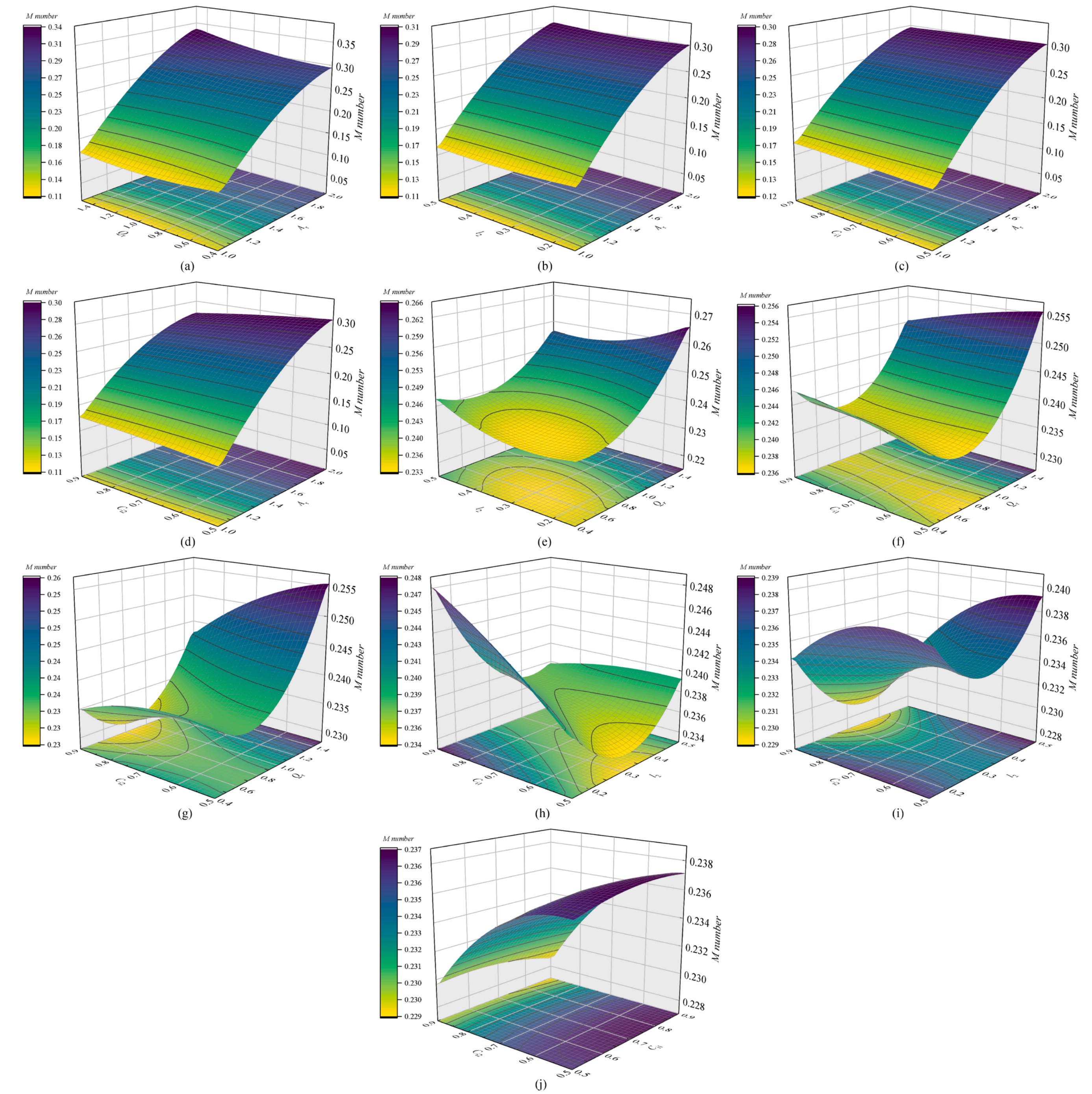

Fig. 19. The contours of RSM analysis on *M number*: (a) Effect of interaction A_r and Q_r $_{on\ on}$ $_{M\ number}$(l_2 = 0.425 m, C_{i1} = 0.5 wt%, C_{i2} =0.5 wt%), (b) Effect of interaction A_r and l_2 on *M number* (Q_r = 0.8, C_{i1} =0.5 wt%, C_{i2} =0.5 wt%), (c) Effect of interaction A_r and C_{i1} on *M number* (A_r = 1.5, l_2 = 0.425 m, C_{i2} = 0.5 wt%), (d) Effect of interaction A_r and C_{i2} on *M number* (A_r = 1.5, l_2 = 0.425 m, C_{i1} = 0.5 wt%), (e) Effect of interaction Q_r and l_2 on *M number* (A_r = 1.5, C_{i1} =0.5 wt%, C_{i2} = 0.5 wt%), (f) Effect of interaction Q_r and C_{i1} on *M number* (A_r = 1.5, l_2 = 0.425 m, C_{i2} = 0.5 wt%), (g) Effect of interaction Q_r and C_{i2} on *M number* (A_r = 1.5, l_2 = 0.425 m, C_{i1} =0.5 wt%), (h) Effect of interaction l_2 and C_{i1} on *M number* (A_r = 1.5, Q_r =0.8, C_{i2} = 0.5 wt%), (i) Effect of interaction l_2 and C_{i2} on *M number* (A_r = 1.5, Q_r =0.8, C_{i1} =0.5 wt%) and (j) Effect of interaction C_{i1} and C_{i2} on *M number* (A_r = 1.5, Q_r =0.8, l_2 = 0.425 m).

observed that the $\overline{LE}_{ave}$ exhibits a descending trend as *M number* increases. Consequently, the optimal point D as the solution set is calculated. The point D with population size of 300 and iteration times of 500 is finally selected as the optimal solution based on the running time of MATLAB program. The design variables corresponding to this optimal solution are $A_r = 1.684$, $Q_r = 1.5$, $l_2 = 0.499$ m, $C_{i1} = 0.5$ wt% and $C_{i2} =$ 0.5 wt%, respectively. Specifically, the *M number* is 0.2926 and $\overline{LE}_{ave}$ is 5.3246. The *M number* and $\overline{LE}_{ave}$ are higher by 40.10 % and 54.47 % than those of the original design, respectively.

5. Conclusion

In order to enhance the mixing efficiency and system stability of static mixer, an improved S-Type static mixer (S-Type-b) is developed. The S-Type-b static mixer optimizes the internal flow structure by removing the side baffle. The structure simplifies the fluid path, reduces

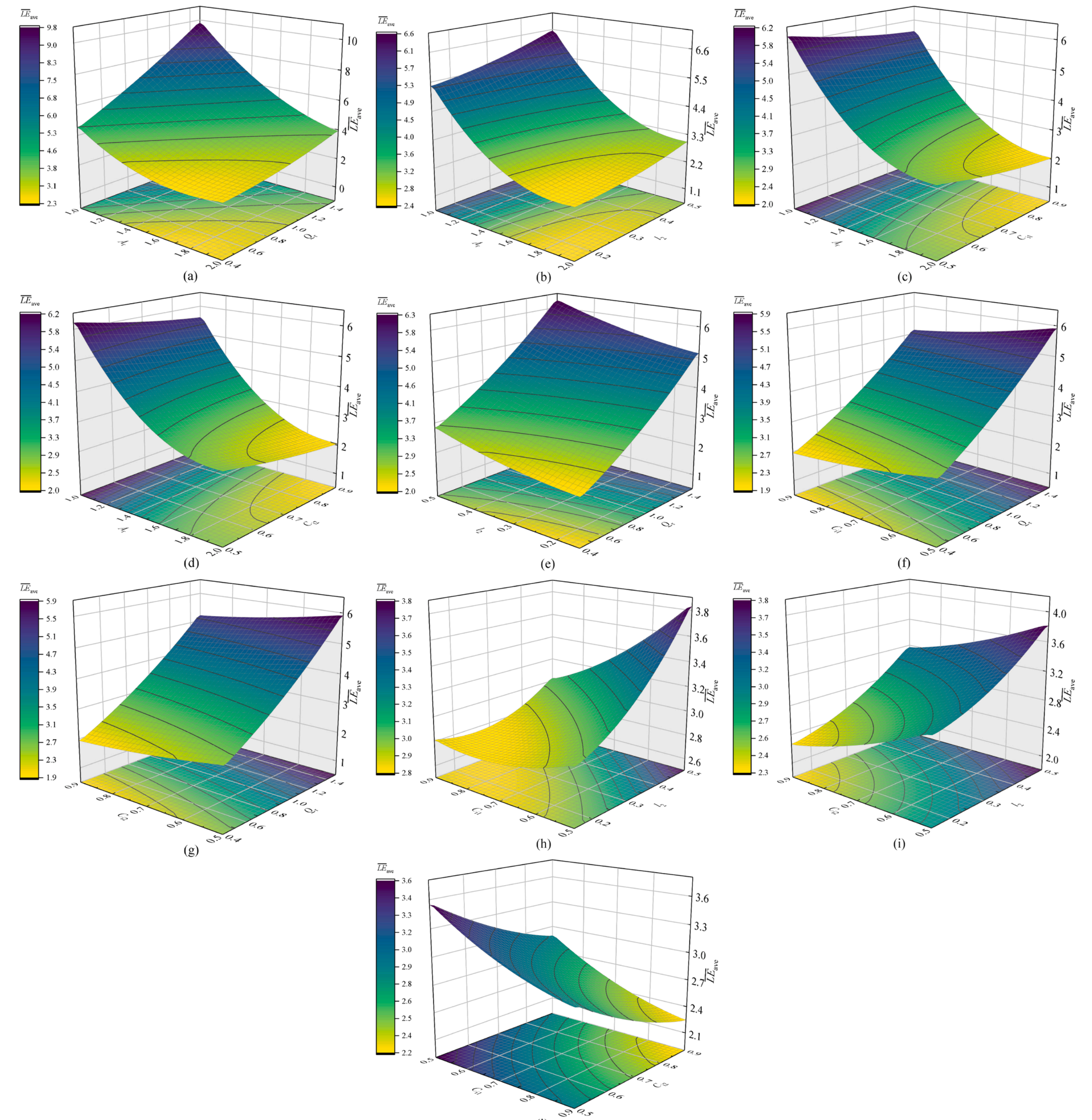

Fig. 20. The contours of RSM analysis on $\overline{LE}_{ave}$: (a) Effect of interaction A_r and Q_r on the $\overline{LE}_{ave}$ ($l_2 = 0.425$ m, C_{i1} =0.5 wt%, C_{i2} =0.5 wt%), (b) Effect of interaction A_r and l_2 on the $\overline{LE}_{ave}$ ($Q_r = 0.8$, C_{i1} =0.5 wt%, C_{i2} =0.5 wt%), (c) Effect of interaction A_r and C_{i1} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, $l_2 = 0.425$ m, C_{i2} =0.5 wt%), (d) Effect of interaction A_r and C_{i2} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, $l_2 = 0.425$ m, C_{i1} =0.5 wt%), (e) Effect of interaction Q_r and l_2 on the $\overline{LE}_{ave}$ ($A_r = 1.5$, C_{i1} =0.5 wt%, C_{i2} =0.5 wt%), (f) Effect of interaction Q_r and C_{i1} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, $l_2 = 0.425$ m, C_{i2} =0.5 wt%), (g) Effect of interaction Q_r and C_{i2} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, $l_2 = 0.425$ m, C_{i1} =0.5 wt %), (h) Effect of interaction l_2 and C_{i1} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, Q_r =0.8, C_{i2} =0.5 wt%), (i) Effect of interaction l_2 and C_{i2} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, Q_r =0.8, C_{i1} =0.5 wt%) and (j) Effect of interaction C_{i1} and C_{i2} on the $\overline{LE}_{ave}$ ($A_r = 1.5$, Q_r =0.8, $l_2 = 0.425$ m).

the local resistance and minimizes fluid retention area compared with the S-Type. This innovative design effectively increases the fluid shear rate and maintains flow stability. The effects of mixing and chaotic characteristics of four static mixers under the different flow rates and cross sections are comprehensively studied by numerical simulation. The S-Type-b static mixer is further optimized using RSM and NSGA-II. The conclusions are summarized as follows:

(1) The improved S-Type-b mixer reduces the resistance and enhances the fluid shear rate during mixing, resulting in obviously higher distribution mixing uniformity than other mixers, the Z_f of S-Type-b static mixer decreases by 5.60–17.86 %, 10.60–21.58 %

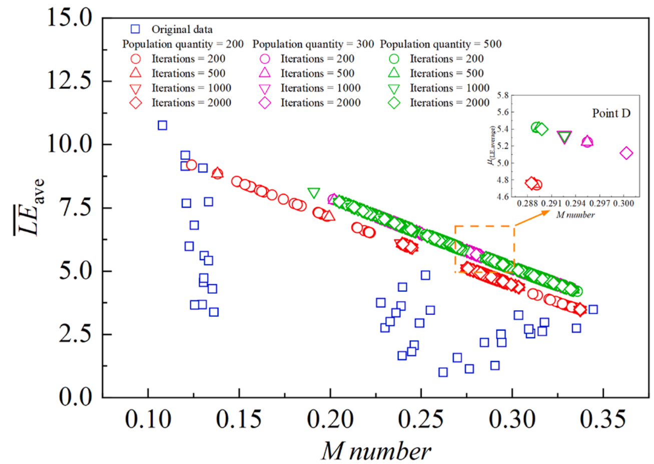

Fig. 21. The distribution of the Pareto-optimal solutions.

and 13.25–24.98 % compared with the Kenics, Komax and S-Type at Q_{i1} ranging from 0.25 to 2.5 L/min.

(2) The *MI* of S-Type-b static mixer at $z/l = 8$ is higher by 0.29–2.78 %, 1.36–4.02 %, and 0.18–4.43 % than that of the Kenics, Komax and S-Type mixers. Furthermore, the striation distribution gradually becomes thinner and reaches a uniform state after the fifth mixing elements.

(3) The innovative design of S-Type-b mixer could achieve better mixing performance based on the given energy consumption for non-Newtonian fluids. The *Pc* of the S-Type-b mixer is reduced by up to 51.5 % compared to the S-Type mixer. The *M number* of S-Type-b static mixer is 10.13–18.29 %, 10.82–22.26 %, and 20.39–27.35 % higher than that of Kenics, Komax, and S-Type, respectively. The dispersion mixing effect of S-Type-b with the average α ranging from 0.56 to 0.62 is higher by 12.45–18.60 % than that of S-Type.

(4) The S-Type-b static mixer presents higher LE_{ave} and lower LE_{var} under all flow rates, which demonstrates that it can induce stronger chaotic mixing performance in the mixing process with stable flow structure. The LE_{ave} slightly decreases after reaching the peak value at $z/l = 7$ for higher flow rates because of the weakening of chaotic behaviors. Meanwhile, LE_{var} gradually increases suggesting that flow instability is intensified with the increase of flow rates.

(5) The effects of A_r, Q_r and l_2 on *M number* and $\overline{LE}_{ave}$ are remarkable through RSM analysis. The optimal design parameters (A_r = 1.684, l_2 = 0.499 m, Q_r = 1.5, C_{i1} = 0.5 wt% and C_{i2} = 0.5 wt%) achieve a balance between efficient mixing and chaotic complexity when the ideal *M number* of 0.2926 and $\overline{LE}_{ave}$ of 5.3246 are obtained.

CRediT authorship contribution statement

Yanfang Yu: Writing – original draft, Supervision, Investigation, Conceptualization. **Wen Li:** Software, Formal analysis. **Huibo Meng:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Kexin Xiang:** Visualization, Investigation. **Deao Li:** Visualization. **Ruiyu Xia:** Investigation. **Shunyao Yu:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cep.2024.110112.

Data availability

Data will be made available on request.

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